



LED TV

Chassis : UNV72

Model : UN24M4500AF
UN28M4500AF

SERVICE Manual

LED TV



UN**M4500AF

Contents

1. Precautions
2. Product specifications
3. Disassembly and Reassembly
4. Troubleshooting
5. Wiring Diagram

Contents

1. Precautions	1-1
1-1. Safety Precautions	1-1
1-1-1. Warnings	1-1
1-1-2. Servicing the LED TV	1-1
1-1-3. Fire and Shock Hazard	1-1
1-1-4. Product Safety Notices	1-2
1-2. Servicing Precautions	1-3
1-2-1. General Servicing Precautions	1-3
1-3. Static Electricity Precautions	1-4
1-4. Installation Precautions	1-5
2. Product Specifications.....	2-1
2-1. Product information	2-1
2-2. Product specification	2-2
2-2-1. Detailed Specifications.....	2-2
2-2-2. Specifications.....	2-13
2-3. Accessories	2-14
3. Disassembly and Reassembly	3-1
3-1. Disassembly and Reassembly	3-1
4. Troubleshooting	4-1
4-1. Troubleshooting	4-1
4-2. How to Check Fault Symptom	4-2
4-2-1. Power	4-2
4-2-2. Main	4-5
4-2-3. Video	4-7
4-2-4. Audio	4-9
4-2-5. Network	4-10
4-2-6. WiFi Module	4-11
4-2-7. FUNCTION/IR Control	4-12
4-3. Factory Mode Adjustments	4-13
4-3-1. Detail Factory Option	4-13
4-3-2. Entering Factory Mode.....	4-15
4-4. White Balance	4-22
4-4-1. Calibration	4-22
4-4-2. Service Adjustment	4-22
4-5. Software Upgrade.....	4-24
4-5-1. By USB	4-24
4-5-2. By Online	4-24
4-5-3. Alternative Software (Backup).....	4-24

5. Wiring Diagram 5-1
5-1. Wiring Diagram 5-1
5-2. Connector 5-2
ANNEX. Exploded View & Part List [UN24M4500AFXZA WA01]ANNEX-1
1-1. Exploded ViewANNEX-1
1-1-1. Parts ListANNEX-1
2-1. Electrical Parts ListANNEX-2
ANNEX. Exploded View & Part List [UN28M4500AFXZA BA01]ANNEX-1
1-1. Exploded ViewANNEX-1
1-1-1. Parts ListANNEX-1
2-1. Electrical Parts ListANNEX-2



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1. Precautions

1-1. Safety Precautions

Follow these safety, servicing and ESD precautions to prevent damage and to protect against potential hazards such as electrical shock.

1-1-1. Warnings



For continued safety, do not attempt to modify the circuit board.
Disconnect the AC power and DC power jack before servicing.

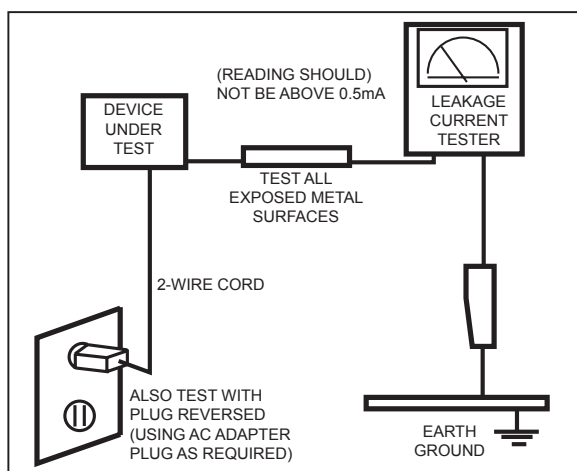
1-1-2. Servicing the LED TV

1. When servicing the LED TV, Disconnect the AC line cord from the AC outlet.
2. It is essential that service technicians have an accurate voltage meter available at all times. Check the calibration of this meter periodically.

1-1-3. Fire and Shock Hazard

Before returning the monitor to the user, perform the following safety checks:

1. Inspect each lead dress to make certain that the leads are not pinched or that hardware is not lodged between the chassis and other metal parts in the monitor.
2. Inspect all protective devices such as nonmetallic control knobs, insulating materials, cabinet backs, adjustment and compartment covers or shields, isolation resistor/capacitor networks, mechanical insulators, etc.
3. Leakage Current Hot Check:



Do not use an isolation transformer during this test.

Use a leakage current tester or a metering system that complies with American National Standards Institute (ANSI C101.1, Leakage Current for Appliances), and Underwriters Laboratories (UL Publication UL1410, 59.7).

4. With the unit completely reassembled, plug the AC line cord directly into a 120V AC outlet. With the unit's AC switch first in the ON position and then OFF, measure the current between a known earth ground (metal water pipe, conduit, etc.) and all exposed metal parts, including: metal cabinets, screwheads and control shafts.
The current measured should not exceed 0.5 milliamp.
Reverse the power-plug prongs in the AC outlet and repeat the test.

1-1-4. Product Safety Notices

Some electrical and mechanical parts have special safetyrelated characteristics which are often not evident from visual inspection. The protection they give may not be obtained by replacing them with components rated for higher voltage, wattage, etc. Parts that have special safety characteristics are identified by  on schematics and parts lists. A substitute replacement that does not have the same safety characteristics as the recommended replacement part might create shock, fire and/or other hazards. Product safety is under review continuously and new instructions are issued whenever appropriate.

1-2. Servicing Precautions



An electrolytic capacitor installed with the wrong polarity might explode.



Before servicing units covered by this service manual, read and follow the Safety Precautions section of this manual.



If unforeseen circumstances create conflict between the following servicing precautions and any of the safety precautions, always follow the safety precautions.

1-2-1. General Servicing Precautions

1. Always unplug the unit's AC power cord from the AC power source and disconnect the DC Power Jack before attempting to: (a) remove or reinstall any component or assembly, (b) disconnect PCB plugs or connectors, (c) connect a test component in parallel with an electrolytic capacitor.
2. Some components are raised above the printed circuit board for safety. An insulation tube or tape is sometimes used. The internal wiring is sometimes clamped to prevent contact with thermally hot components. Reinstall all such elements to their original position.
3. After servicing, always check that the screws, components and wiring have been correctly reinstalled. Make sure that the area around the serviced part has not been damaged.
4. Check the insulation between the blades of the AC plug and accessible conductive parts (examples: metal panels, input terminals and earphone jacks).
5. Insulation Checking Procedure: Disconnect the power cord from the AC source and turn the power switch ON. Connect an insulation resistance meter (500 V) to the blades of the AC plug. The insulation resistance between each blade of the AC plug and accessible conductive parts (see above) should be greater than 1 megohm.
6. Always connect a test instrument's ground lead to the instrument chassis ground before connecting the positive lead; always remove the instrument's ground lead last.

1-3. Static Electricity Precautions

Some semiconductor (solid state) devices can be easily damaged by static electricity. Such components are commonly called Electrostatically Sensitive Devices (ESD). Examples of typical ESD are integrated circuits and some field-effect transistors. The following techniques will reduce the incidence of component damage caused by static electricity.

1. Immediately before handling any semiconductor components or assemblies, drain the electrostatic charge from your body by touching a known earth ground. Alternatively, wear a discharging wrist-strap device. To avoid a shock hazard, be sure to remove the wrist strap before applying power to the monitor.
2. After removing an ESD-equipped assembly, place it on a conductive surface such as aluminum foil to prevent accumulation of an electrostatic charge.
3. Do not use freon-propelled chemicals. These can generate electrical charges sufficient to damage ESDs.
4. Use only a grounded-tip soldering iron to solder or desolder ESDs.
5. Use only an anti-static solder removal device. Some solder removal devices not classified as "anti-static" can generate electrical charges sufficient to damage ESDs.
6. Do not remove a replacement ESD from its protective package until you are ready to install it. Most replacement ESDs are packaged with leads that are electrically shorted together by conductive foam, aluminum foil or other conductive materials.
7. Immediately before removing the protective material from the leads of a replacement ESD, touch the protective material to the chassis or circuit assembly into which the device will be installed.



Be sure no power is applied to the chassis or circuit and observe all other safety precautions.

8. Minimize body motions when handling unpackaged replacement ESDs. Motions such as brushing clothes together, or lifting your foot from a carpeted floor can generate enough static electricity to damage an ESD.

1-4. Installation Precautions

1. For safety reasons, more than a people are required for carrying the product.
2. Keep the power cord away from any heat emitting devices, as a melted covering may cause fire or electric shock.
3. Do not place the product in areas with poor ventilation such as a bookshelf or closet. The increased internal temperature may cause fire.
4. Bend the external antenna cable when connecting it to the product. This is a measure to protect it from being exposed to moisture. Otherwise, it may cause a fire or electric shock.
5. Make sure to turn the power off and unplug the power cord from the outlet before repositioning the product. Also check the antenna cable or the external connectors if they are fully unplugged. Damage to the cord may cause fire or electric shock.
6. Keep the antenna far away from any high-voltage cables and install it firmly. Contact with the highvoltage cable or the antenna falling over may cause fire or electric shock.
7. When installing the product, leave enough space (0.4m) between the product and the wall for ventilation purposes. A rise in temperature within the product may cause fire.
8. If an equipment is provided with a replaceable battery, and if replacement by an incorrect type could result in an explosion (for example, with some lithium batteries), the following applies:



- Risk of explosion if battery is replaced by an incorrect type dispose of used batteries according to the instructions.
- Do not dispose of batteries in a fire.
- Do not short circuit, disassemble or overheat the batteries.
- Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type.
- Do not be exposed to excessive heat such as sunshine, fire or the like.

2. Product Specifications

2-1. Product information

Model	UN**M4500AF		
Front View	 * W : Width H : High D : Depth		
Detail View			
Color	Front : GLOSSY BLACK, Stand : BLACK		
Dimensions (W x H x D)	24"	With Stand	561.8 x 384.2 x 163.8 mm
		Without Stand	561.8 x 349.1 x 47.9 mm
	28"	With Stand	643.4 x 435.0 x 163.4 mm
		Without Stand	643.4 x 396.5 x 62.9 mm
Weight	24"	With Stand	4.2 kg
		Without Stand	4.0 kg
	28"	With Stand	3.8 kg
		Without Stand	3.4 kg

2-2. Product specification

2-2-1. Detailed Specifications

 NOTE Design and specifications are subject to change without prior notice.		
Item		UN24M4500AFXZA
General Information	Product	LED
	Cabinet Basic Code	U24MD2
	Series	4
	Country	UNITED STATES
	Tools Support	N/A
	Platform(TV)	SoC KANT-S
Display	Inch	24
	Real Inch	23.6
	Q Display	N/A
	Resolution	1,366 x 768
	Screen Curvature	N/A
	10bit Support	N/A
	Bendable Panel	N/A
	Ultra Black	N/A
Video	Picture Engine	HyperReal
	Motion Rate	60
	PQI (Picture Quality Index)	400
	HDR (High Dynamic Range)	N/A
	Q Contrast Ultimate	N/A
	Q Contrast	N/A
	Dynamic Contrast Ratio	Mega Contrast
	Micro Dimming	M/D Pro
	Precision Black (Local Dimming)	N/A
	Q Color	N/A
	Dynamic Crystal Color	N/A
	Active Crystal Color	N/A
	Wide Color Enhancer (Plus)	Yes
	PurColor	Yes
	Q Everyview	N/A
	Auto Depth Enhancer	N/A

Item		UN24M4500AFXZA
Video	Contrast Enhancer	Yes
	Auto Motion Plus	Yes
	Film Mode	Yes
	Picture	WSS,HDMI Black Level
	Response Time	8ms
	Viewing Angle (H/V)	178/178
	Natural Mode Support	Yes
	Peak Illuminator	N/A
	Quantum Dot Color	N/A
Audio	Dolby Digital Plus	Yes
	DTS Codec	Yes
	Sound Output (RMS)	10W(5W+5W)
	Speaker Type	2CH(Down Firing + Base Reflex)
	Woofer	N/A
	Main Speaker Output (W)	10W(5W+5W)
	Woofer Speaker Output (W)	N/A
	Twitter Speaker Output (W)	N/A
	Multiroom Link	N/A
	Bluetooth Audio	N/A
Smart Service	Smart TV Type	Smart
	Voice Interaction	N/A
	TV Plus	N/A
	Web Browser	Yes
	One app Support	N/A
	Smart View	N/A
VESA Standard	Screw Size	M4
	Screw depth	6.3 ~ 9.3
	VESA Spec	75.0 × 75.0
Smart Feature	TV to Mobile - Mirroring	Yes
	Mobile to TV - Mirroring, DLNA	Yes
	360 Video Player	N/A
	360 Camera Support	N/A
	Family Square with S-Cloud	N/A
	Together play	Yes
	Easy Setup	N/A
	App Casting	Yes

2. Product specifications

Item		UN24M4500AFXZA
Smart Feature	Wireless TV On - Samsung WOL	Yes
	Wired TV On - Samsung WOL	Yes
	Bluetooth Low Energy	N/A
	RVU	N/A
	WiFi Direct	Yes
	CVP-2	N/A
	TV as Hub Support	N/A
	TV as Things Support	N/A
	IoT Client Application	N/A
Localization	S-Share	N/A
	Dongle Compatibility (3G / LTE / WiFi)	N/A
	Wireless Copy	N/A
	Sound Mirroring	N/A
	Analog Clean View	N/A
	Cricket Prediction	N/A
	Sports Mode	N/A
	Senior mode	N/A
	Clean View	N/A
	Family TV 2.0	N/A
	Local Cinema Mode	N/A
	All Care Protection	N/A
Feature	Decor Mode Picture Engine	N/A
	Decor Mode	N/A
	Low Power (Decor Mode)	N/A
	Bluetooth Speaker (Frame)	N/A
	Motion Detection (Frame)	N/A
	Intelligent Picture and Sound - Dynamic Media	N/A
	Intelligent Picture and Sound- IP Streaming	N/A
	Instant On	Yes
	Processor	Quad-core
	SCSA Support	N/A
	Accessibility	Voice Guide(US English)/ Enlarge/ High Contrast/ Learn TV Remote(US English)
	Digital Clean View	Yes
	Ultra Clean View	Yes

Item		UN24M4500AFXZA
Feature	Auto Channel Search	Yes
	Auto Power Off	Yes
	Caption (Subtitle)	Yes
	ConnectShare™ (HDD)	Yes
	ConnectShare™ (USB 2.0)	Yes
	ConnectShare™ Transfer	N/A
	Embedded POP	Yes
	EPG	Yes
	Extended PVR	N/A
	Game Mode	Yes
	OSD Language	ENG,EST,FIN,FRA,DEU,GRE,HUN,ITA,LAT,LTU, NOR,POL,POR,ROM,SER,SLK,SPA,SWE,BUL,C RO,CZE,DAN,DUT,SLV,ALB,MKD,BOS,KOR
	Picture-In-Picture	N/A
	BT HID Support	N/A
	USB HID Support	Yes
	Teletext (TTX)	N/A
	Time Shift	N/A
	One Connect	N/A
	V-Chip	Yes
	MBR Support	N/A
	IPv6 Support	Yes(US)
	Gigabit	N/A
Additional Feature	-	-
System	Digital Broadcasting	ATSC / Clear QAM
	DTV Sound System	Dolby
	Analog Tuner	Yes
	Analog Tuner Type	NTSC 3.58
	2 Tuner	N/A
	CI (Common Interface)	N/A
	Data Broadcasting	N/A
	ATV Sound System	M
	Tuner Vendor & Model	(SoluM) DTVS20EH7A
	TV Key	N/A
Core Component	DDR SDRAM	Hynix
	Flash Memory	EMMC Toshiba
	Serial Flash Memory	WINBOND

2. Product specifications

Item		UN24M4500AFXZA
Core Component	HDMI Switch	N/A
	Display Device Vender	Innolux
Connectivity	HDMI	2
	Resolution	1920 x 1080p 60
	DVI Support Port	N/A
	MHL Support Port	N/A
	USB	1
	Port 1 Type	2.0
	Port 2 Type	N/A
	Port 3 Type	N/A
	Port 4 Type	N/A
	Port 5 Type	N/A
	Component In (Y/Pb/Pr)	1
	Composite In (AV)	1(Common Use for Component Y)
	Ethernet (LAN)	1
	Audio Out (Mini Jack / LR)	N/A
	Digital Audio Out (Optical)	1
	RF In (Terrestrial / Cable input / Satellite Input)	1/1(Common Use for Terrestrial)/0
	Ex-Link (RS-232C)	N/A
	CI Slot	N/A
	Monitor Output	N/A
	DVI	
	Resolution	N/A
	D-Sub Resolution	N/A N/A
	HDMI A / Return Ch. Support	Yes
	HDMI Quick Switch	Yes
	Wireless LAN Adapter Support	N/A
	Wireless LAN Built-in	Yes
	Anynet+ (HDMI-CEC)	Yes
Design	Design	North America H4500 re-use
	Bezel Type	NNB(12mm)
	Slim Type	Slim 1 (49mm)
	Front Color	Glossy Black
	Light Effect (Deco)	N/A
	Stand Type	Square (Black)

Item		UN24M4500AFXZA
Design	Swivel (Left/Right)	N/A
Eco	Energy Efficiency Class	Yes
	Eco Sensor	N/A
Item		UN28M4500AFXZA
General Information	Product	LED
	Cabinet Basic Code	U28MD2
	Series	4
	Country	UNITED STATES
	Tools Support	N/A
	Platform(TV)	SoC KANT-S
Display	Inch	28
	Real Inch	27.5
	Real cm	69
	Q Display	N/A
	Resolution	1,366 x 768
	Screen Curvature	N/A
	10bit Support	N/A
	Bendable Panel	N/A
	Ultra Black	N/A
Video	Picture Engine	HyperReal
	Motion Rate	60
	PQI (Picture Quality Index)	400
	HDR (High Dynamic Range)	N/A
	Q Contrast Ultimate	N/A
	Q Contrast	N/A
	Dynamic Contrast Ratio	Mega Contrast
	Micro Dimming	M/D Pro
	Precision Black (Local Dimming)	N/A
	Q Color	N/A
	Dynamic Crystal Color	N/A
	Active Crystal Color	N/A
	Wide Color Enhancer (Plus)	Yes
	PurColor	Yes
	Q Everyview	N/A
	Auto Depth Enhancer	N/A

2. Product specifications

Item		UN28M4500AFXZA
Video	Contrast Enhancer	Yes
	Auto Motion Plus	Yes
	Film Mode	Yes
	Picture	WSS,HDMI Black Level
	Response Time	8ms
	Viewing Angle (H/V)	178/178
	Natural Mode Support	Yes
	Peak Illuminator	N/A
	Quantum Dot Color	N/A
Audio	Dolby Digital Plus	Yes
	DTS Codec	Yes
	Sound Output (RMS)	10W(5W+5W)
	Speaker Type	2CH(Down Firing + Base Reflex)
	Woofer	N/A
	Main Speaker Output (W)	Yes
	Woofer Speaker Output (W)	Yes
	Twitter Speaker Output (W)	10W(5W+5W)
	Multiroom Link	N/A
	Bluetooth Audio	N/A
Smart Service	Smart TV Type	Smart
	Voice Interaction	N/A
	TV Plus	N/A
	Web Browser	Yes
	One app Support	N/A
	Smart View	N/A
VESA Standard	Screw Size	M4
	Screw depth	16.5~19.5
	VESA Spec	100.0 x 100.0
Smart Feature	TV to Mobile - Mirroring	Yes
	Mobile to TV - Mirroring, DLNA	Yes
	360 Video Player	N/A
	360 Camera Support	N/A
	Family Square with S-Cloud	N/A
	Together play	Yes
	Easy Setup	N/A
	App Casting	Yes

Item		UN28M4500AFXZA
Smart Feature	Wireless TV On - Samsung WOL	Yes
	Wired TV On - Samsung WOL	Yes
	Bluetooth Low Energy	N/A
	RVU	N/A
	WiFi Direct	Yes
	CVP-2	N/A
	TV as Hub Support	N/A
	TV as Things Support	N/A
	IoT Client Application	N/A
Localization	S-Share	N/A
	Dongle Compatibility (3G / LTE / WiFi)	N/A
	Wireless Copy	N/A
	Sound Mirroring	N/A
	Analog Clean View	N/A
	Cricket Prediction	N/A
	Sports Mode	N/A
	Senior mode	N/A
	Clean View	N/A
	Family TV 2.0	N/A
	Local Cinema Mode	N/A
	All Care Protection	N/A
Feature	Decor Mode Picture Engine	N/A
	Decor Mode	N/A
	Low Power (Decor Mode)	N/A
	Bluetooth Speaker (Frame)	N/A
	Motion Detection (Frame)	N/A
	Intelligent Picture and Sound - Dynamic Media	N/A
	Intelligent Picture and Sound- IP Streaming	N/A
	Instant On	Yes
	Processor	Quad-core
	SCSA Support	N/A
	Accessibility	Voice Guide(US English)/ Enlarge/ High Contrast/ Learn TV Remote(US English)/ Multi-output Audio
	Digital Clean View	Yes
	Ultra Clean View	Yes

2. Product specifications

Item		UN28M4500AFXZA
Feature	Auto Channel Search	Yes
	Auto Power Off	Yes
	Caption (Subtitle)	Yes
	ConnectShare™ (HDD)	Yes
	ConnectShare™ (USB 2.0)	Yes
	ConnectShare™ Transfer	N/A
	Embedded POP	Yes
	EPG	Yes
	Extended PVR	N/A
	Game Mode	Yes
	OSD Language	ENG,EST,FIN,FRA,DEU,GRE,HUN,ITA,LAT,LTU, NOR,POL,POR,ROM,SER,SLK,SPA,SWE,BUL,C RO,CZE,DAN,DUT,SLV,ALB,MKD,BOS,KOR
	Picture-In-Picture	N/A
	BT HID Support	N/A
	USB HID Support	Yes
	Teletext (TTX)	N/A
	Time Shift	N/A
	One Connect	N/A
	V-Chip	Yes
	MBR Support	N/A
	IPv6 Support	Yes(US)
	Gigabit	N/A
Additional Feature	-	-
System	Digital Broadcasting	ATSC / Clear QAM
	DTV Sound System	Dolby
	Analog Tuner	Yes
	Analog Tuner Type	NTSC 3.58
	2 Tuner	N/A
	CI (Common Interface)	N/A
	Data Broadcasting	N/A
	ATV Sound System	M
	Tuner Vendor & Model	(SoluM) DTVS20EH7A
	TV Key	N/A
Core Component	DDR SDRAM	Hynix
	Flash Memory	EMMC Toshiba
	Serial Flash Memory	WINBOND
	HDMI Switch	N/A

Item		UN28M4500AFXZA
Core Component	Display Device Vender	BOE
Connectivity	HDMI	2
	Resolution	N/A
	DVI Support Port	N/A
	MHL Support Port	N/A
	USB	1
	Port 1 Type	2.0
	Port 2 Type	N/A
	Port 3 Type	N/A
	Port 4 Type	N/A
	Port 5 Type	N/A
	Component In (Y/Pb/Pr)	1
	Composite In (AV)	1(Common Use for Component Y)
	Ethernet (LAN)	1
	Audio Out (Mini Jack / LR)	N/A
	Digital Audio Out (Optical)	1
	RF In (Terrestrial / Cable input / Satellite Input)	1/1(Common Use for Terrestrial)/0
	Ex-Link (RS-232C)	N/A
	CI Slot	N/A
	Monitor Output	N/A
	DVI	
	Resolution	N/A
	D-Sub Resolution	N/A N/A
	HDMI A / Return Ch. Support	Yes
	HDMI Quick Switch	Yes
	Wireless LAN Adapter Support	N/A
	Wireless LAN Built-in	Yes
	Anynet+ (HDMI-CEC)	Yes
Design	Design	North America H4500 re-use
	Bezel Type	NNB(12mm)
	Slim Type	Slim 1 (49mm)
	Front Color	Glossy Black
	Light Effect (Deco)	N/A
	Stand Type	Square (Black)
	Swivel (Left/Right)	N/A

2. Product specifications

Item		UN28M4500AFXZA
Eco	Eco Sensor	N/A
	Mercury Content (mg)	0.0
	Lead Presence	Yes

2-2-2. Specifications

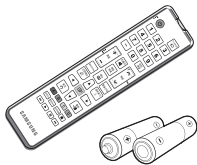
■ Specifications

Model	UN**M4500AF	
Item	Description	
Screen Size (Diagonal)	24 inches	28 inches
LCD Panel	FHD 60Hz	
Display Resolution	1,366 x 768	
Input Signal	Analog 0.7 Vp-p $\pm$ 5% positive at 75 Ω , internally terminated	
Input Sync Signal	H/V Separate, TTL, P. or N.	
Environmental Considerations	Operating Temperature: 50°F ~ 104°F (10°C ~ 40°C) Operating Humidity: 10% ~ 80%, non-condensing Storage Temperature: -4°F ~ 113°F (-20°C ~ 45°C) Storage Humidity: 5% ~ 95%, non-condensing	
AC Power Voltage & Frequency	AC110-120V 50/60Hz	
Sound (Output)	10W (5W X 2)	

2-3. Accessories

 **NOTE**

- The items' colors and shapes may vary depending on the model.
- Cables not included in the package contents can be purchased separately.
- The part code for some accessories may differ depending on your region.

Product	Description	Code. No		Remark
	Remote Control & Batteries (AAA x 2)	BN59-01267A		-
		4301-000121		
	Power Cord	3903-001117		
	Manual Users	24"	BN68-08058F	
		28"	BN68-08058H	
	HOLDER-STAND CABLE	28"	BN61-08370A	

3. Disassembly and Reassembly

This section of the service manual describes the disassembly and reassembly procedures for the LED TV.



This LED TV contains electrostatically sensitive devices. Use caution when handling these components.

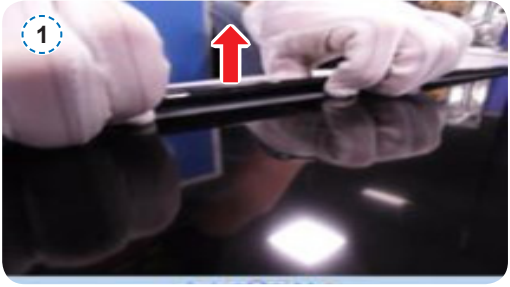
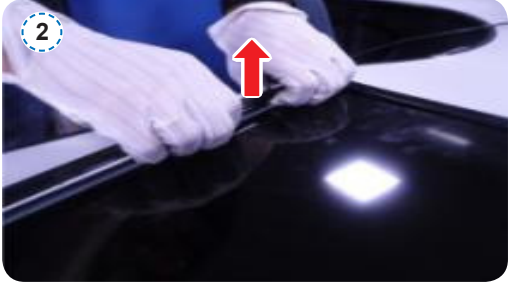
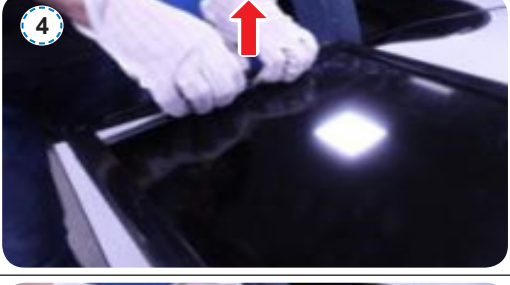
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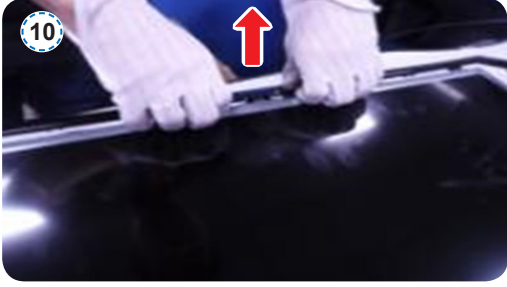


1. Disconnect the LED TV from the power source before disassembly.
2. Follow these directions carefully; never use metal instruments to pry apart the cabinet.
3. If there is no additional coment, it is same for all inches.

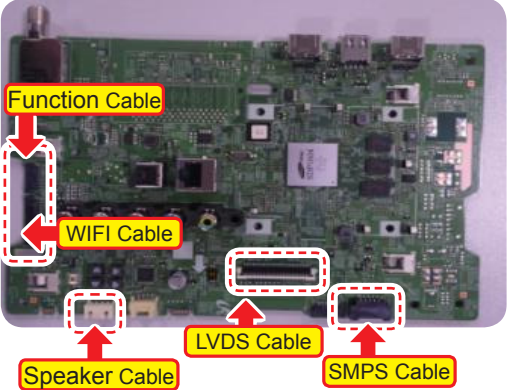
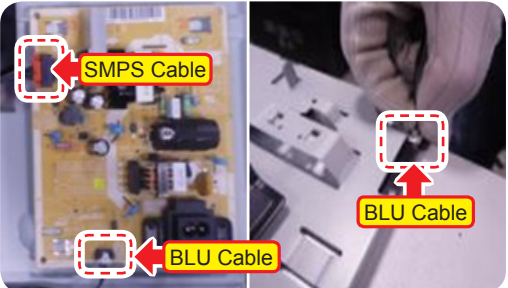
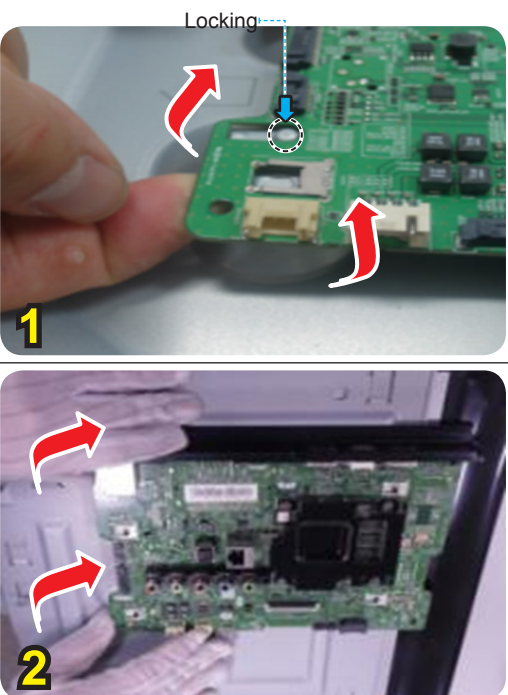
Description	Picture Description	Screws
1 Place TV face down on cushioned table.		
2 Remove the DC VSS(A) and 3 Screws from the ASSY STAND P-BASE.		 6003-001782
3 Grip the Cover Front is rotated by 45 degrees outwards aand pulls to release the cover mounting clips.	 	

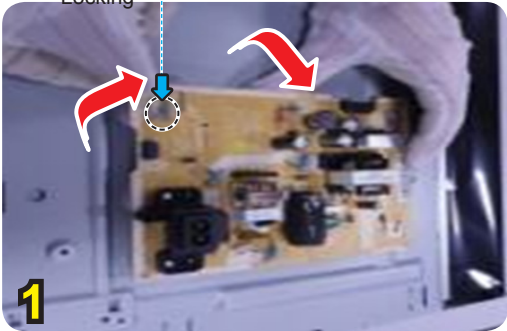
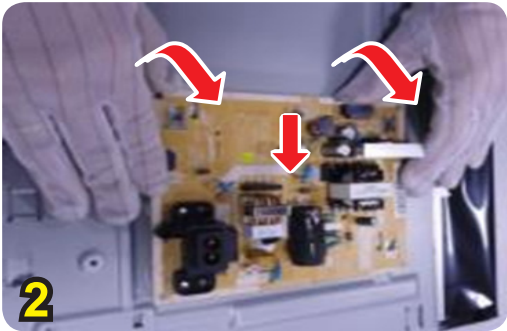
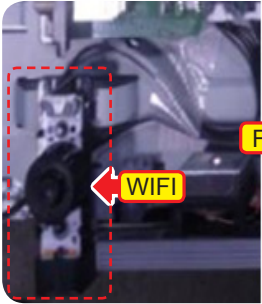
3. Disassembly and Reassemble

Description	Picture Description	Screws
Open the Top hook (Middle)		
Open the Top hook (Right side)		
Open the Top hook (Left side)		
Open the Side hook (Right side)		
Open the Side hook (Right side)		
Open the Side hook (Left side)		

Description	Picture Description	Screws
Open the Side hook (Left side)	 A close-up photograph showing a person's hands in white gloves pulling a metal hook on the left side of a black surface. A red arrow points upwards, indicating the direction of movement. A blue circle with the number 7 is in the top-left corner.	
Open the Bottom hook (Left side)	 A close-up photograph showing a person's hands in white gloves pulling a metal hook on the bottom-left side of a black surface. A red arrow points upwards, indicating the direction of movement. A blue circle with the number 8 is in the top-left corner.	
Open the Bottom hook (Right side)	 A close-up photograph showing a person's hands in white gloves pulling a metal hook on the bottom-right side of a black surface. A red arrow points upwards, indicating the direction of movement. A blue circle with the number 9 is in the top-left corner.	
Open the Bottom hook (Middle)	 A close-up photograph showing a person's hands in white gloves pulling a metal hook in the middle of the bottom edge of a black surface. A red arrow points upwards, indicating the direction of movement. A blue circle with the number 10 is in the top-left corner.	
4 Remove the Cover Front and Cover Rear	 A photograph showing the internal components of a device with the front and rear covers removed. The internal components, including a green circuit board and various connectors, are visible. A person's hand is visible on the left side of the device.	

3. Disassembly and Reassemble

Description	Picture Description	Screws
		
5 Remove the Function Cable, Speaker Cable, SMPS Cable, WIFI Cable and LVDS Cable.		
6 Remove the SMPS Cable and BLU Cable.		
7 Removing Main Board. - Gently lift up (Bottom Right corner) to release the lock. - Use both hands to hold the board and slide UP to release the board.		

Description	Picture Description	Screws
8 Removing SMPS Board. - Gently lift up (Bottom Left corner) to release the lock. - Use both hands to hold the board and slide UP to release.	Locking 	
		
9 Remove the Wifi, Speaker and Function.		
	 	
10 Remove Assy Bracket.		

3. Disassembly and Reassemble

Description	Picture Description	Screws
11 Completed disassembly.		

 **NOTE**

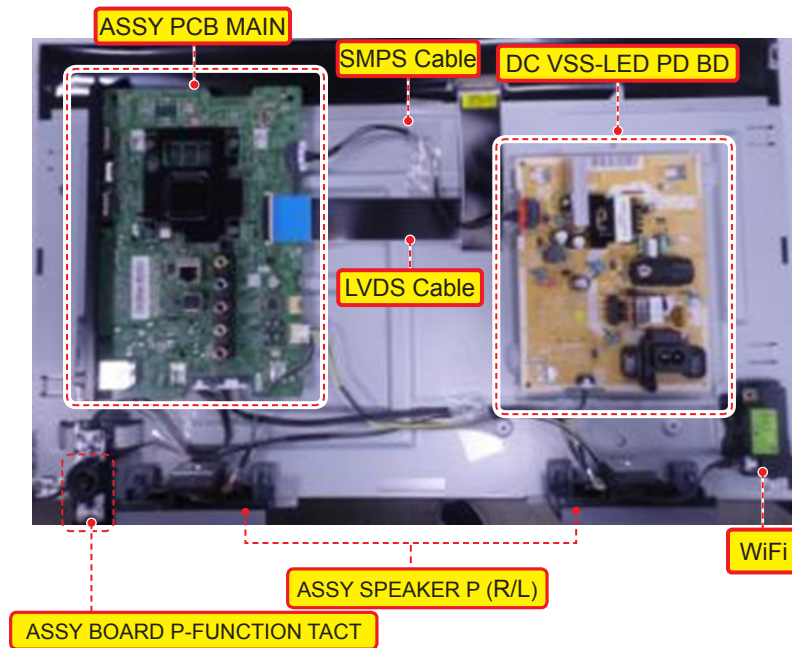
Reassembly procedures are in the reverse order of disassembly procedures.

4. Troubleshooting

4-1. Troubleshooting

■ Previous Check

1. Check the various cable connections first.
 - Check to see if there is a burnt or damaged cable.
 - Check to see if there is a disconnected or loose cable connection.
 - Check to see if the cables are connected according to the connection diagram.
2. Check the power input to the ASSY PCB MAIN.



3. Check the power in & output between SMPS(DC VSS-LED PD BD) & Main Board, Main Board & Panel, IP & Panel

4-2. How to Check Fault Symptom

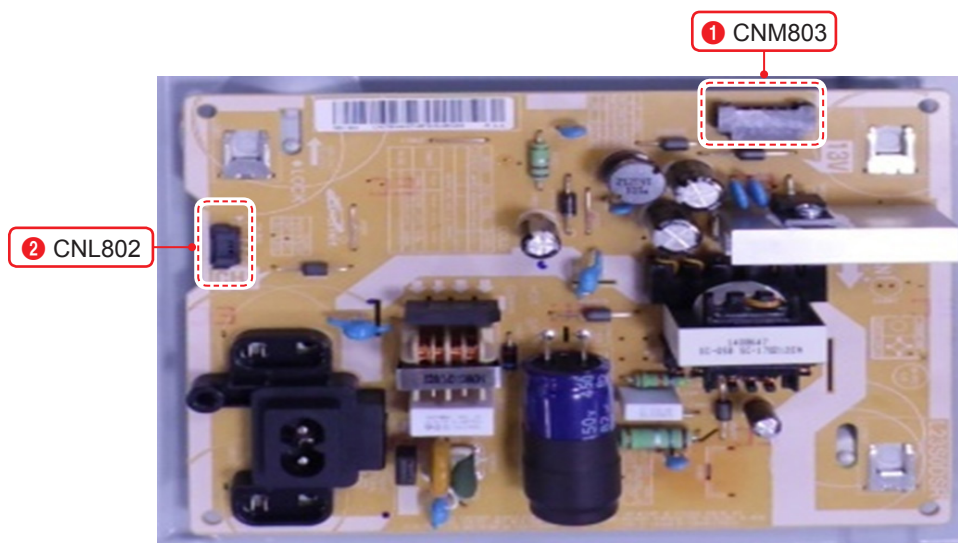
4-2-1. Power

■ TV POWER STANDBY

1. TV in Standby
 - √ **Standby LED Indicator**
2. If Not Lit:
 - √ AC 120Vac Line
3. If missing:
 - √ 120Vac Source and Power Cord
4. If OK:
 - √ Resistance on SMPS **Fuse** after first removing AC power cord.
5. If fuses are open replace SMPS.
6. If fuses are OK:
 - √ **Standby: A13V** (Always On) to Main Board. Should all be approx. 9 **VDC**
7. If any missing remove the SMPS connector to Main Board .
 - √ Standby A13V again for 9VDC.
 - If OK replace **Main Board**.
 - If still missing replace **SMPS**.

■ SMPS POWER

SMPS_Front



Detail

1	To Main Board CNM803			
2	OD_UD	1	A13V	
4	BLU_PWM	3	A13V	
6	Power On/Off	5	A13V	
8	GND	7	A13V	
10	GND	9	GND	

2	To Panel LEDS CNL802	
1	ILED	
2	ILED	
3	VLED	
4	VLED	
5	ILED	
6	ILED	

Testing

1. **A13V** Standby to Main Board : Approx 9Vdc.
2. **Power_On/Off** : 3.3Vdc Power ON/OFF.
3. **BLU_PWM** : .9V~3.3Vdc also labeled "**BLU On/Off**" Backlight On/Off & Backlight (PWM signal) Level Control.

■ BACKLIGHTS



1. Activate Backlights Test:

- Disconnect Lead Cable from Main Board to Power Supply. (CNM803)
- ✓ TV Screen for active backlight LEDs.

2. If NO BACKLIGHTS

- ✓ Minus (Control) & Plus pins (Supply) on the Panel Connector voltages to the Panel.
- If no pin voltages replace SMPS.

3. If BACKLIGHTS ON BUT PANEL SECTION(S) OFF

- ✓ The Supply Drive + pins. All should measure same.
- If a + pin measures higher voltage, a string(s) of LEDs are likely open.
- Remove Panel connector and verify same open backlight voltage condition. **Replace Panel.**

4. If a + pin measures low voltage disconnect connector to panel

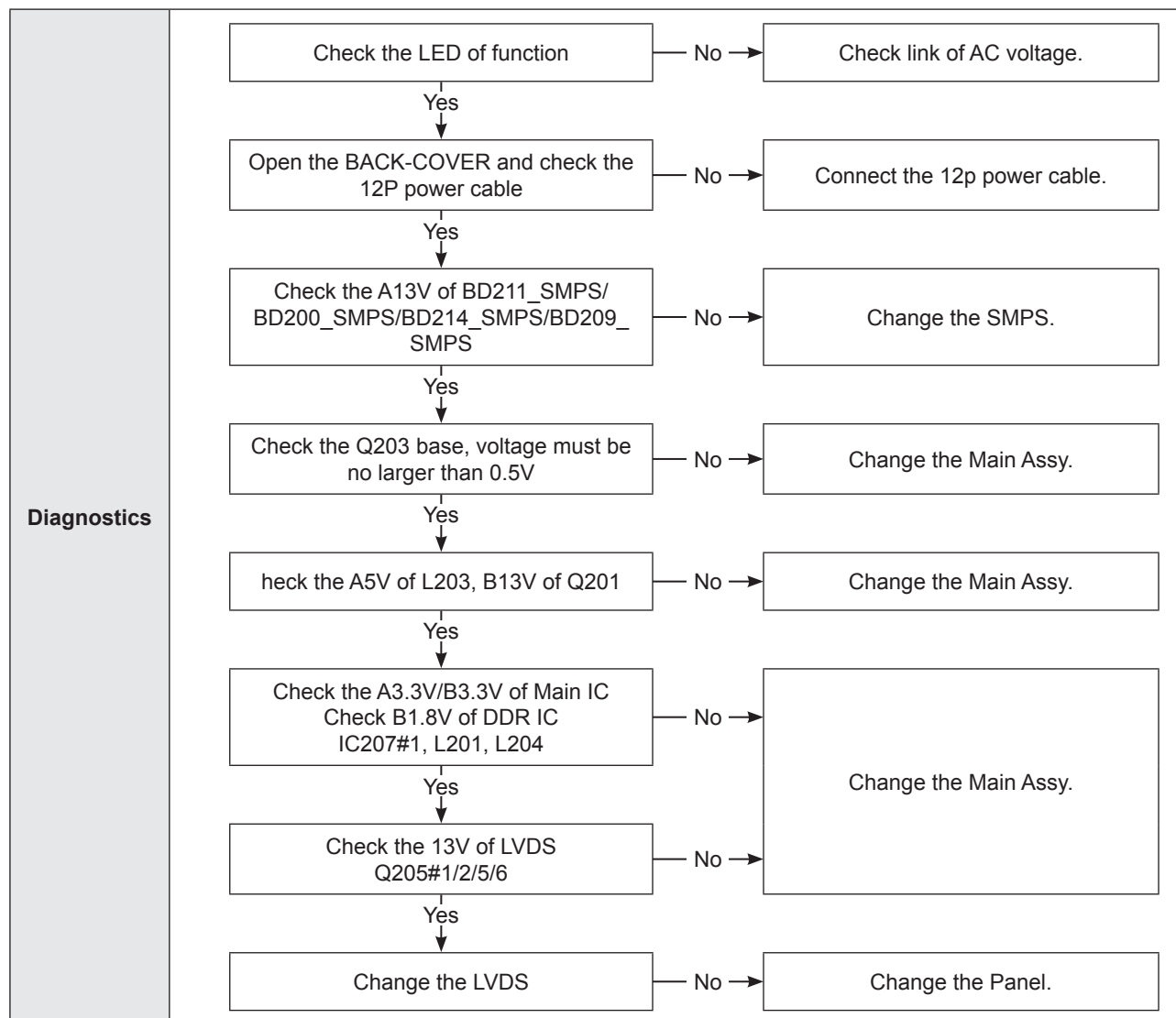
- ✓ The Low Plus Pin Voltage again.
- If it stays low the **SMPS** is defective, if it goes high, defective **Panel**.

For BACKLIGHT DIMMING PROBLEMS:

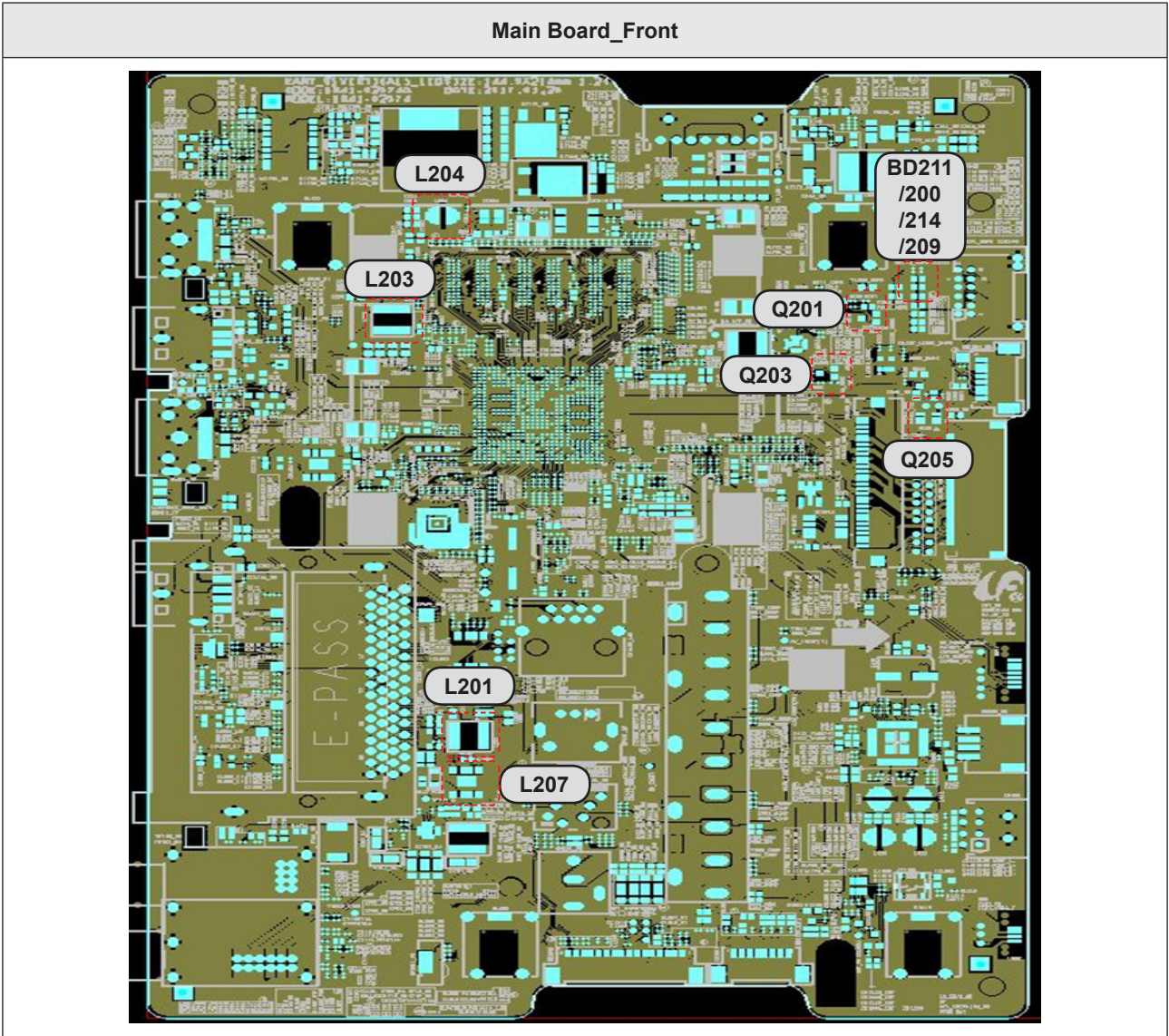
- Go to **Menu > Picture > Backlight** and vary level (0 – 20)
- If no backlight changes observed:
 - ✓ Panel Connector pin voltages and BLU_PWM voltages (CNL802) while changing backlight level.
- If Panel voltages don't change, and BLU_PWM changes, replace **SMPS**.
- If BLU_PWM doesn't change replace **Main/T-CON Board**.

4-2-2. Main

■ No power & No picture



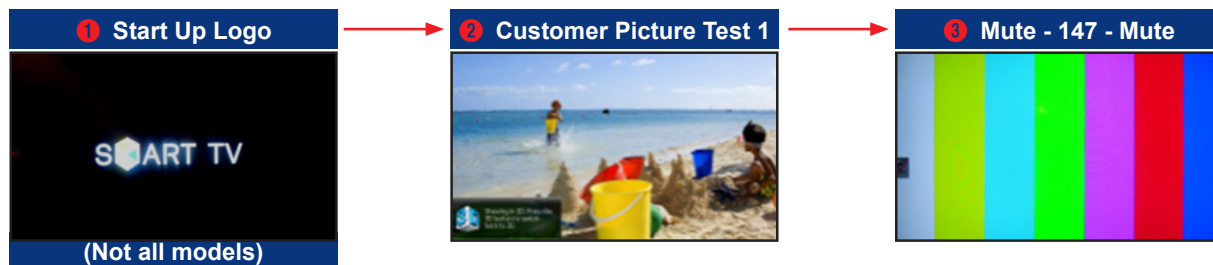
■ Location of Parts



4-2-3. Video

■ MAIN

1. Main Board Section



Video Operation : **Generated on Main Section.**

1. If OK:

- ✓ Source & Input Cables.
- ✓ Other inputs.
- ✓ One Connect Cable/Box.

2. If Noisy:

- ✓ Panel (defective).

■ PANEL

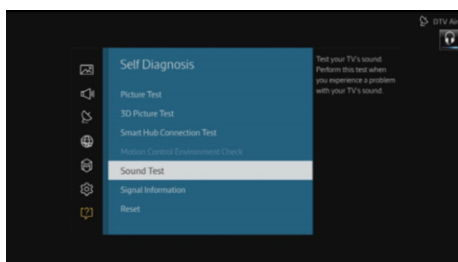


1. If Noisy:

- ✓ Panel (defective).

4-2-4. Audio

AUDIO



- **No TV Sound**
 - ✓ Menu → Audio → Speaker Settings set to **TV Speaker**
- **Noisy / Distorted TV Audio**
 - ✓ Customer Menu → Support → **Sound Test**
- **If **Sound Test** FAILS : (Missing / Noisy Audio)**
 - ✓ Speakers (compare resistance/quality)
 - ✓ Main Board
 - Compare audio level out to speakers with multi meter.
 - ✓ Replace defective Speakers or Main Board or Cable.
- **IF **Sound Test** OK :**
 - ✓ Audio Source & External Cables.
 - ✓ Other Inputs.
 - ✓ With external Audio Generator (device or App).
- **Optical Digital Out Errors**
 - ✓ Red light from Optical Digital Out.
- **No HDMI Audio**
 - ✓ Source / HDMI Cable
 - Swap with other HDMI Inputs/Sources.
 - Perform **EDID Write** in Factory Mode (Can restore missing HDMI Audio).
 - ✓ Bulletins and Latest firmware on TV.
 - If not restored replace One Connect Box.
 - Check Audio Format PCM/Dolby based on external Receiver.
- **ARC Issues**
 - ✓ HDMI Cable is input to the ARC Designated HDMI port.
 - ✓ ARC (HDMI Control) is enabled on the external Receiver.
- **Bluetooth Audio "**Sound Share**" Connection Issues**
 - ✓ Sound Bar is in TV Mode.
 - To Connect, Press & Hold Play Button until Sound Bar pairing mode begins.

4-2-5. Network

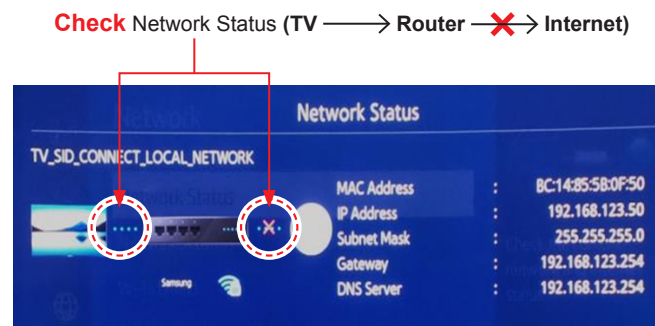


■ TV to Router "Failure"



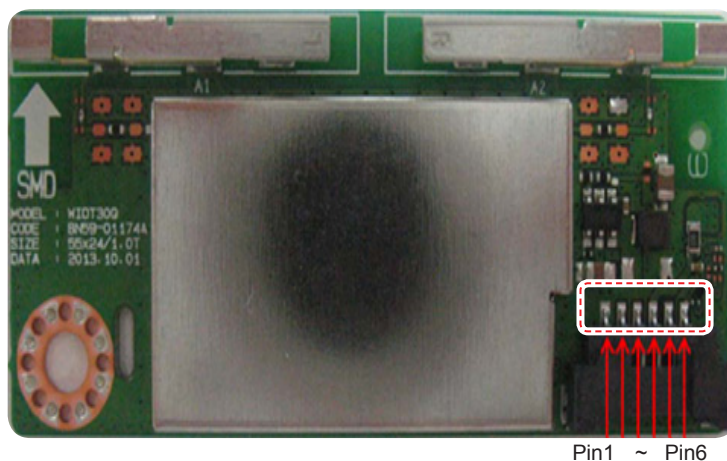
- **Wired & Wireless MAC Address** in Customer Support Menu.
 - ✓ **No Wired MAC Address:** Replace Main Board.
 - ✓ **No Wireless MAC Address:** Module cabling & voltages from Main Board.
 - ✓ If operating voltages are OK but signal missing.
 - Replace WiFi Module (WiFi/Bluetooth Module).
- Proper **security passcode**
 - ✓ Check Wi-Fi signal strength at TV (use WiFi Analyzer or similar App).
 - Try another source (Hot spot or Test Router).
 - ✓ Check related Bulletins.
 - ✓ Check **Factory Mode** → **SVC** → **Info** → **WiFi Error Count** (replace module for high error counts).

■ Router to Network "Failure"



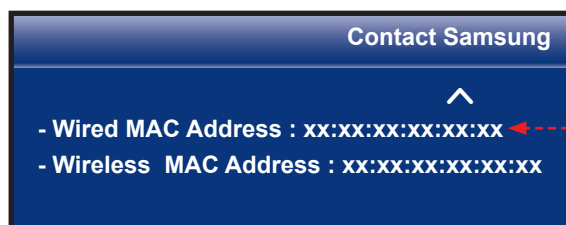
- Instruct the customer the TV has proper connection to the router and is likely OK.
 - ✓ Check other devices using network are OK. If they test OK this does not mean the TV should be working.
- Try another source (Hotspot) to test/show TV Network operation.

4-2-6. WiFi Module

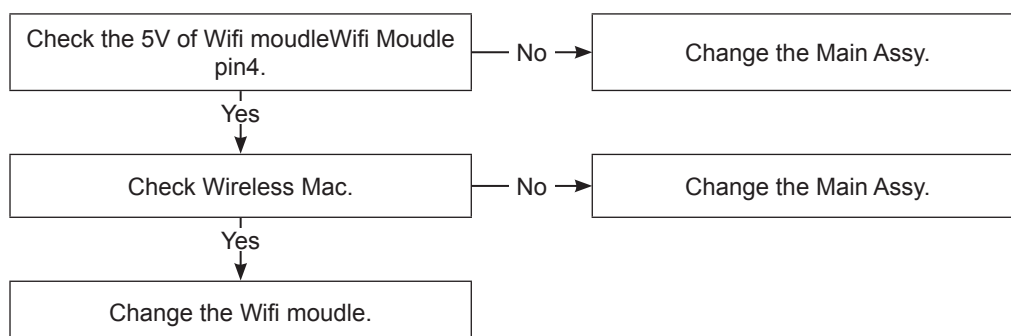


Pins #	Pin Name	Description	Type
1	GND	Ground	G
2	USB DP	USB interface	I/O
3	USB DN	USB interface	I/O
4	B+_5V	+5V DC power supply for SWL-R55	V
5	Wake up	Sleep/OFF to ON	I/O
6	NC	-	-

Go to [Menu](#) → [Support](#) → [Contact Samsung](#)

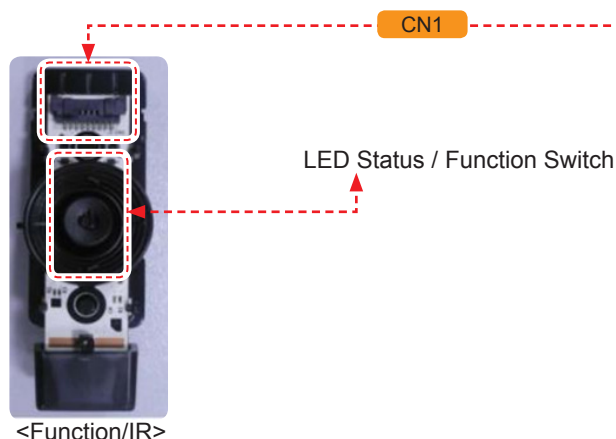


Missing or Error : Replace **Main Board**.



4-2-7. FUNCTION/IR Control

CN1 (FUNCTION/IR)			
1	IR GND	2	LED
3	KEY2	4	KEY1
5	SDA 3.3 Vdc (effective DC)	6	SCL 3.3 Vdc (effective DC)
7	3.3V	8	GND
9	IR		



Direction	KEY1	KEY2
Center	0.00 ~ 0.30V	3.00 ~ 3.30V
CH+	3.00 ~ 3.30V	0.00 ~ 0.30V
CH-	3.00 ~ 3.30V	0.50 ~ 1.10V
Vol+	3.00 ~ 3.30V	1.30 ~ 1.95V
Vol-	3.00 ~ 3.30V	2.15 ~ 2.80V
Normal	3.00 ~ 3.30V	3.00 ~ 3.30V

1. TV in Standby

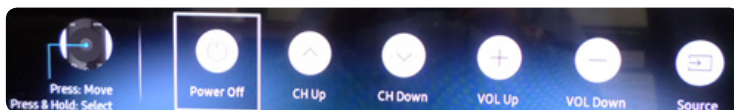
✓ **LED Status**

2. If LED is OFF

- ✓ LED 1.7Vdc (pin 8) and VCC for 3.3Vdc (pin 3)
 - If missing suspect **Function Assy/Cable Assy/Main Assy**.

3. If LED is ON

- ✓ Switch Operation activates on screen display.



<On Screen Selections with Function Control>

- If missing:
 - ✓ **Key _Input1 Pin 6** change to 0V with a command.
 - If wrong voltage or no change:
 - ✓ Switch for stuck or miss-operation.
4. Check **IR** operation with Standard Remote command changes. (3.3V to 2.5V effective DC)
5. **SDA, SCL** for effective 3.3Vdc (after power on)
- If missing suspect **Function Assy/Cable Assy/Main Assy**.

4-3. Factory Mode Adjustments

4-3-1. Detail Factory Option



NOTE

If you replace the main board with new one, please change the factory option as well.
The options you must change are "Type".

■ UN24M4500AFXZA

• PANEL / SMPS / MAIN Information

Multi Bom	PANEL			SMPS		MAIN	
WA01	Vendor	CHIMEI		Vendor	POWER	ASSY CHASSIS	ASSY PCB MAIN
	Code	Spec	Type	Code	Spec		
	BN07-01378B	V236BJ1-LE2	24D6AH0E	BN44-00746F	L23S0DSR_MPN	BN91-19087C	BN94-12232A

• Factory Option

Local Set	BOM Model	Front Color	S/W Model
US	4500	N/A	UM4500

■ UN28M4500AFXZA

• PANEL / SMPS / MAIN Information

Multi Bom	PANEL			SMPS, ADAPTOR		MAIN	
BA01	Vendor	BOE		Vendor	DYREL	ASSY CHASSIS	ASSY PCB MAIN
	Code	Spec	Type	Code	Spec		
	BN95-04265A	CY-HM028AGEV1V/H	28B6AH0S	BN44-00835A	A4819_FDY	BN91-19087B	BN94-12232B

Multi Bom	PANEL			SMPS, ADAPTOR		MAIN	
BB02	Vendor	BOE		Vendor	DYREL	ASSY CHASSIS	ASSY PCB MAIN
	Code	Spec	Type	Code	Spec		
	BN95-04265A	CY-HM028AGEV1V/H	28B6AH0S	BN44-00835D	A4819_MSM	BN91-19087B	BN94-12232B

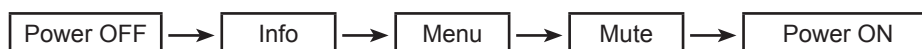
• Factory Option

Local Set	BOM Model	Front Color	S/W Model
US	4500	N/A	UM4500

4-3-2. Entering Factory Mode

To enter 'Service Mode' Press the remote -control keys in this sequence :

- If you do not have Factory remote control



- If you have Factory remote control



- If you don't have Factory remote control, can't control some menus.

Option	T-KTSAKUC_-xxxx T-KTSINTV_-xxxx TIZEN-3.0-MAIN2017-KantS-RELEASE_2017****.*
Control	
Debug	
SVC	E-Manual : KTSATSCL-*.*.*
ADC/WB	Blaster Version :Not Support
Advanced	E-POP Version :KANTSFD-**.**.*
	EDID SUCCESS
	HDCP SUCCESS
	CALIB : AV/COMP/PC/HDMI/
	Option : xxxx,xxxx,xxxx,NONE
	FRC-[Kant-S Vx1][60Hz]
	DIMMING-[GLOBAL]
	TCON Version: ----
	Model : *****
	Wired MAC SUCCESS
	Wireless MAC SUCCESS
	WIFI Version: *****
	CO Nf/ W/ MO D/ H2 PO AO O S/ N/ RO SC/ SiO WS/ DI/ UO I/ (T)
	NS//
	Factory Data Ver : xxxxx / Fixed Ver : xxxxx
	EERC Version : xxx / WB Ver : 1
	CPLD/LD :N/A
	SmartControl : ****
	Board Info : ----
	Factory Reset In Production : ----
	SID : ----
	Date of purchase : mm/dd/yyyy

■ Factory Remote

1. Power TV ON.
2. Select TV Source.
3. **Info** → **Factory**.
4. Use **MENU** for return.

■ Samsung IR Remote (Limited Operation)

1. TV Power Standby.
2. Press as follows.

• Remote Button : **NTSC** : **MUTE** → **1** → **8** → **2** → **POWER ON**

PAL : **INFO** → **MENU** → **MUTE** → **POWER ON**

- **Option** (must set Option Bytes when replacing Main Board.)
- Option → **Factory Reset** (returns TV to out of box condition. Does not reset Apps.)
- SVC → **Test Patterns**
- SVC → Info → **ER Count** (Important to check for errors.)
 - Resets to 0 with Factory Reset.

First Screen Appearing in Factory Mode

- Ex. SAMPLE Model

Home	Updates	Exit
Option		
Control		
Debug		
SVC		
ADC/WB		
Advanced		

T-KTMAKUC-0604.10
 T-KTMINTV-0045
 TIZEN-3.0-MAIN2017-KantM-RELEASE_20170104.1 (Debug)
 Main SPI version : 0
 OCM SPI version : 0
 BT Version : BLUETOOTH-VER-0653
 E-Manual : KTMAISCL-0.0.5
 Blaster Version : A61206-U61001-170201
 E-POP Version : KANTMUD-12.28.3
 EDID SUCCESS
 HDCP SUCCESS
 CALIB : AV / COMP / PC / HDMI /
 Option : 55A1AU0SMJS,8000,NONE
 FRC-[KANT-M USIT][120Hz][HW:0x0F]
 DIMMING-[EDGE-8X1][03]
 TCON-[KANT-M] FW[801B] DATA[M55ABU0F15]
 Model : UN55MU8000
 Wired MAC SUCCESS
 Wireless MAC SUCCESS
 WiFi Version : 4.5.30.015.046.fw4646
 CO NFO WO M / D / H2 PO AO O S / N / RO SC / SIOWS / DI / UO I / (T)
 NS//
 Factory Data Ver : 17063 / Fixed Ver : 1706
 EERC Version : 82 / WB Ver : 1
 CPLD/LD : N/A
 SmartControl : A0700200
 Board Info : 2016/12/19/PVR/1/BN41-02570A
 Factory Reset In Production : ----
 SID : ?
 Date of purchase : --/--/----

QUICK TIPS:-
 * Set value of HV Flip from shell prompt without launching factory app. For detail check Updates window
 * Launch Factory:- "org.tizen.factory"
 * Get value of any item from shell prompt without launching factory app. For detail check Updates window
 * Launch Factory:- "org.tizen.factory"
 * Set value of type localset and model from shell prompt along with factory reset:-
 "launch_app org.tizen.factory type value localset value model value"
 * Use channel up/down to toggle items when at last depth
 * Use arrow right/left to toggle values when at last depth
 * Home button can be used to jump directly to home screen of factory app from any level
 * Exit button can be used to Exit factory app without launching aging app

- Slide details (Ex. SAMPLE Model)

Home	Updates	Exit
Option		
Control		
Debug		
SVC		
ADC/WB		
Advanced		

T-KTSAKUC_-xxxx

T-KTSINTV_-xxxx

TIZEN-3.0-MAIN2017-KantS-RELEASE_2017****.*

E-Manual : KTSATSCL-*.**.*

Blaster Version :Not Support

E-POP Version :KANTSFD-*.**.**.*

EDID SUCCESS

HDCP SUCCESS

CALIB : AV/COMP/PC/HDMI/

Option : xxxx,xxxx,xxxx,NONE

FRC-[Kant-S Vx1][60Hz]

DIMMING-[GLOBAL]

TCON Version: ----

Model : *****

Wired MAC SUCCESS

Wireless MAC SUCCESS

WIFI Version: *****

CO Nf/ W/ MO D/ H2 PO AO O S/ N/ RO SC/ SiC

WS/ DI/ UO I/ (T)

NS//

Factory Data Ver : xxxxx / Fixed Ver : xxxx

EERC Version : xxx / WB Ver : 1

CPLD/LD :N/A

SmartControl : ****

Board Info : ----

Factory Reset In Production : ----

SID : ----

Date of purchase : mm/dd/yyyy

✓ **Testing Items**

✓ Micom Version

✓ Sub Micom

✓ Tizen

✓ E-Manual

✓ Wired MAC Success

✓ Wireless MAC Success

✓ CO Status ("O" Operational)

TESTING

1. Verify SW Versions.
2. Verify Wired MAC Addresses SUCCESS.
3. Verify Wireless MAC Address "SUCCESS".
4. Verify DRM Crt, NF, etc Status is "O".

■ Factory Reset

- Ex. SAMPLE Model

Factory Reset	
Type	24D6AH0E
Local Set	US
SW Model	UM4500
BOM Model	4500
TUNER	AUTO
Ch Table	NONE

Setting Option Bytes

1. Enter Factory Mode with **Service Remote** (required to change all settings) and select **Option** screen.
2. Check Option Byte Table located on **GSPN** ((Fast Tracks or Tips) for latest exact model information. (Do not use the Service Manual).
3. Select each item to change change (**Type**, **Local Set**, **SW Model**, **BOM Model**, **Tuner**, etc.).
4. Additional items are located in **MRT Option** such as Front Color, LVDS Format, Region, etc), if necessary to change.

■ Factory Mode > SVC > SVC Info**SVC Info** (Special Notes)

1. Model code : UN24M4500	
2. Serial Number : Fail	Serial Number is Missing "Fail" when Main Board is replaced.
3. Pre Software Version :	
4. Software Version : T-KTSAKUC_-xxxx	Main Micom SW Version
5. Panel Display Time : xxxx min	Panel Hours
6. Last Mode (Connection) : CATV	
7. Pairing : (null)	Currently Paired Items Pass/Fail : Should Pass
8. SmartHub Used : // Operating	
9. watchdog Count : 0	

■ SVC > Info > **ER Count**

MAIN : SVC, MODE : CATV, RES : NOSIGNAL

WD Count	0	Serdes Error Count	0
AR Count	0	Serdes Reset Count	0
RS Count	0	Serdes WatchDog On/Off	ON
WIFI NO DETECTION COUNT	0	SMPS FET Fail Detect	0
WIFI DETACHMENT COUNT	0	Vcc Fail Count	0
BT NO DETECTION COUNT	0		
BT DETACHMENT COUNT	0		
BT MGT OPEN FAIL COUNT	0		
BT MGT DISCONNECT COUNT	0		
BT TV AUDIO DROP	0		
BT AUDIO TIMER EXP	0		
Camera ER Count			
FRC3D Emergency Reboot On/Off	ON		
FRC3D ER Count	0		
Fan Error Count	0		

■ Factory Mode > Control > EDID

1. Remove ALL **HDMI** connections.
2. Factory Mode → Control → **EDID**. (→ **Enter** Key)

Option	EDID
Control	Sub Option
Debug	Hotel Option
SVC	Shop Option
ADC/WB	Asia Option
Advanced	Sound

3. Select EDID/OFF to ON. (→ **Right Arrow** Key)

EDID ON/OFF	ON
-------------	----

4. Select EDID WRITE ALL. (→ **Enter** Key)

EDID WRITE ALL	Success
----------------	---------

5. Wait to Success. (→ **Right Arrow** Key)

EDID WRITE ALL	Wait
----------------	------

6. Confirm EDID WRITE ALL Success. (→ **Menu** Key)

EDID WRITE ALL	Success
----------------	---------

- Check Important WD Count (hardware failures), AR Count (Apps & software failures), Wi-Fi, BT, HDMI, Camera.
- **EDID** : Use for HDMI issues by removing all HDMI connectors, Turning EDID OFF to ON, then selecting EDID write .
- Verify EDID Write Success status.

4-4. White Balance

4-4-1. Calibration

1. Into the Factory Mode.
2. Select **SVC** Menu.
3. Select **ADC/WB** menu.
4. Select **ADC** menu.

Option
Control
Debug
SVC
ADC/WB
Advanced

ADC

AV Calibration
Comp Calibration
PC Calibration
HDMI Calibration

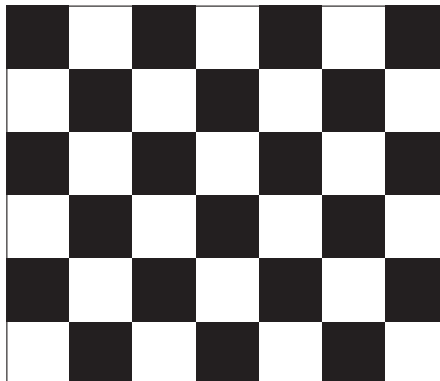
4-4-2. Service Adjustment

You must perform Calibration in the Lattice Pattern before adjusting the White Balance.

■ Color Calibration

- Adjust Specification

Source	Setting Mode	Pattern	Use Equipment
HDMI	1280 x 720@60 Hz	Pattern #24 (Chess Pattern)	CA210 & Master MSPG925 Generator



(Chess Pattern)

- Use other equipment only after comparing the result with that of the Master equipment.

Input mode	Calibration	Pattern
CVBS IN (Model_#1)	Perform in NTSC B&W Pattern #24	Lattice
Component IN (Model_#6)	Perform in 720p B&W Pattern #24	Lattice
PC Analog IN (Model_#21)	Perform in VESA XGA (1024x768) B&W Pattern #24	Lattice
HDMI IN	Perform in 720p B&W Pattern #24	Lattice

■ Method of Color Calibration (AV)

1. Apply the NTSC Lattice (N0. 3) pattern signal to the AV IN 1 port.
2. Press the Source key to switch to "AV1" mode.
3. Enter Service mode.
4. Select the "ADC" menu.
5. Select the "AV Calibration" menu.
6. In "AV Calibration Off" status, press the "▶" key to perform Calibration.
7. When Calibration is complete, it returns to the high-level menu.
8. You can see the change of the "AV Calibration" status from Failure to Success.

■ Method of Color Calibration (Component)

1. Apply the 720p Lattice (N0. 6) pattern signal to the Component IN 1 port.
2. Press the Source key to switch to "Component1" mode.
3. Enter Service mode.
4. Select the "ADC" menu.
5. Select the "Comp Calibration" menu.
6. In "Comp Calibration Off" status, press the "▶" key to perform Calibration.
7. When Calibration is complete, it returns to the high-level menu.
8. You can see the change of the "Comp Calibration" status from Failure to Success.

■ Method of Color Calibration (PC)

1. Apply the VESA XGA Lattice (N0. 21) pattern signal to the PC IN port.
2. Press the Source key to switch to "PC" mode.
3. Enter Service mode.
4. Select the "ADC" menu.
5. Select the "PC Calibration" menu.
6. In "PC Calibration Off" status, press the "▶" key to perform Calibration.
7. When Calibration is complete, it returns to the high-level menu.
8. You can see the change of the "PC Calibration" status from Failure to Success.

■ Method of Color Calibration (HDMI)

1. Apply the 720p Lattice (N0. 6) pattern signal to the HDMI1/DVI IN port.
2. Press the Source key to switch to "HDMI1" mode.
3. Enter Service mode.
4. Select the "ADC" menu.
5. Select the "HDMI Calibration" menu.
6. In "HDMI Calibration Off" status, press the "▶" key to perform Calibration.
7. When Calibration is complete, it returns to the high-level menu.
8. You can see the change of the "HDMI Calibration" status from Failure to Success.

4-5. Software Upgrade

Samsung may offer upgrades for the TV's firmware in the future. These upgrades can be performed via the TV when it is connected to the Internet, or by downloading the new firmware from samsung.com to a USB memory device.

- Alternative Software (Backup) shows The previous version that will be replaced.
- Software is represented as 'Year/Month/Day_Version'. The more recent the date, the newer the software version. Installing the latest version is recommended.



4-5-1. By USB

Insert a USB drive containing the firmware upgrade downloaded from samsung.com into the TV. Please be careful to not disconnect the power or remove the USB drive while upgrades are being applied. The TV will turn off and turn on automatically after completing the firmware upgrade. Please check the firmware version after the upgrades are complete (the new version will have a higher number than the older version). When software is upgraded, video and audio settings you have made will return to their default (factory) settings. We recommend you write down your settings so that you can easily reset them after the upgrade.

"MENU → Support → Software Update"

4-5-2. By Online

Upgrades the software using the Internet.

- First, configure your network. For detailed procedures on using the Network Setting, refer to the 'Setting the Network' instructions.
- If The internet connection doesn't operate properly, connection can be broken, please retry downloading. If the problem still happens, download by USB and upgrade.

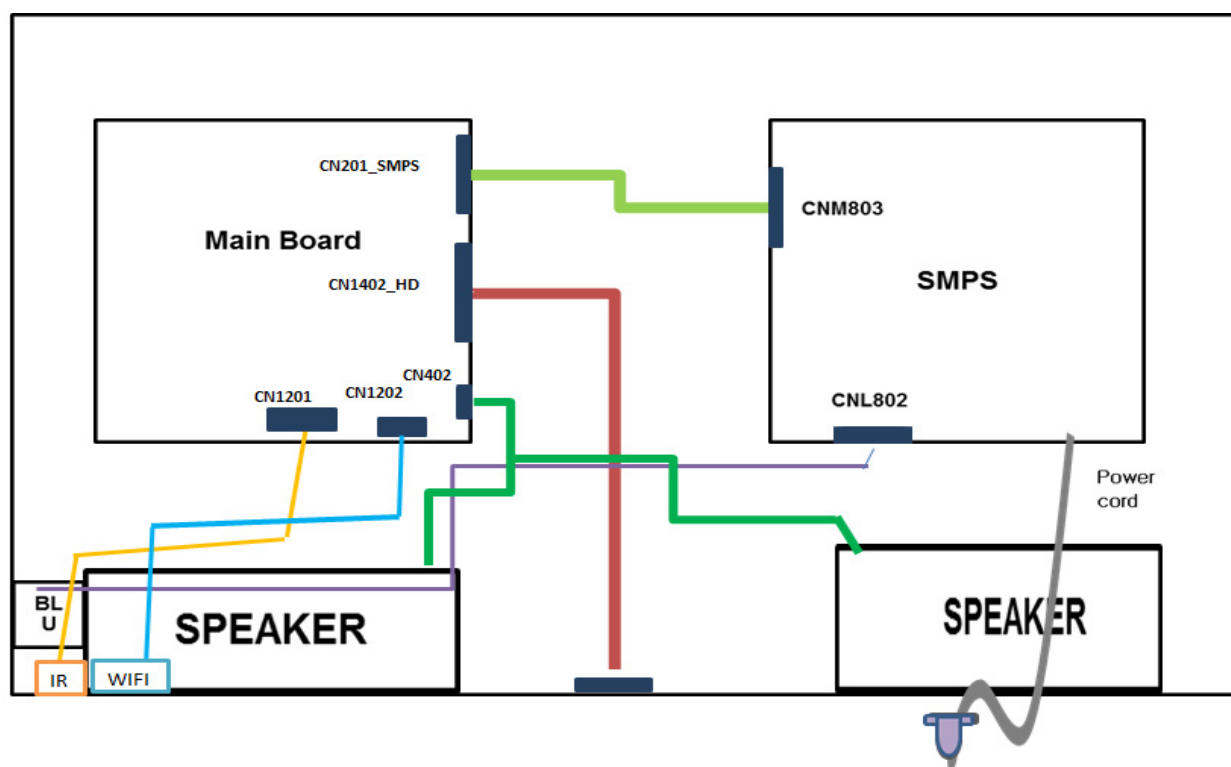
4-5-3. Alternative Software (Backup)

If there is an issue with the new firmware and it is affecting operation, you can change the software to the previous

- If Software was changed, existing Software is displayed.
- you can change current Software to Alternative Software by 'Alternative Software'.

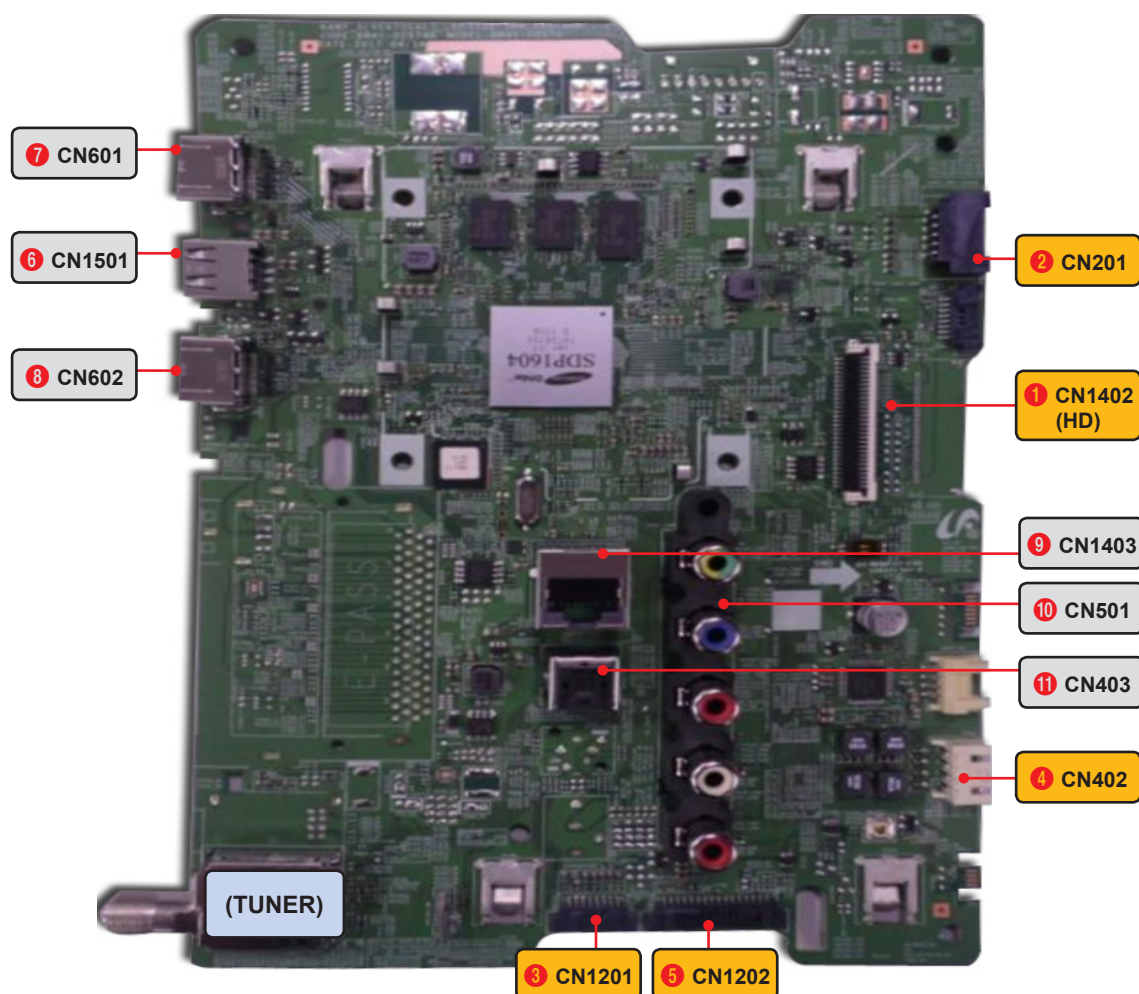
5. Wiring Diagram

5-1. Wiring Diagram



5-2. Connector

■ Main Board



■ Main Board Pin Map

① CN1402_HD			
1	PANEL_13V_PW	16	TXCLK+
2	PANEL_13V_PW	17	TXCLK-
3	PANEL_13V_PW	18	GND
4	PANEL_13V_PW	19	TX2+
5	PANEL_13V_PW	20	TX2-
6	GND	21	GND
7	NC	22	TX1+
8	NC	23	TX1-
9	NC	24	GND
10	NC	25	TX0+
11	NC	26	TX0-
12	GND	27	NC
13	TX3+	28	NC
14	TX3+	29	NC
15	GND	30	NC

② CN201_SMP (to Power board)			
1	GND	7	A13V_PW
2	GND	8	PWM_DIM
3	A13V_PW	9	A13V_PW
4	GND	10	OVD_LEVEL
5	A13V_PW	11	SMPS_FET_FAIL_DETECT
6	SW_POWER_OUT	12	OVD_ON_OFF

③ CN1201 (IR)			
1	IR	5	MSDA_I2C
2	GND	6	KEY_INPUT1
3	A3.3V_PW	7	KEY_INPUT2
4	MSCL_I2C	8	LED_STB_OUT_BS12

4 CN402 (to Speaker)			
1	R+	3	L+
2	R-	4	L-

5 CN1202 (WIFI)			
1	BT_RESET	9	WIFI_PHY_ON
2	BT_POWER_DET	10	GND
3	BT_WAKE	11	WIFI+_USB_WIFI
4	GND	12	WIFI-_USB_WIFI
5	BT-_USB_BT	13	GND
6	BT+_USB_BT	14	A5V_PW
7	GND	15	WIFI_WOL
8	GND	16	WIFI_NRESET

6 CN1501_U1 (USB)			
1	A5V_USB1_PW	3	USB1_DP+_USB
2	USB1_DM-_USB	4	GND

7 CN601_H1 (to HDMI1)			
1	HDMI1_RX2+_HDMI	11	GND
2	GND	12	HDMI1_RX_CLK-_HDMI
3	HDMI1_RX2-_HDMI	13	CEC
4	HDMI1_RX1+_HDMI	14	NC
5	GND	15	HDMI1_SCL_DDC
6	HDMI1_RX1-_HDMI	16	HDMI1_SDA_DDC
7	HDMI1_RX0+_HDMI	17	HDMI1_CABLE_DET
8	GND	18	HDMI1_5V_PW
9	HDMI1_RX0-_HDMI	19	HDMI1_HOT_PLUG
10	HDMI1_RX_CLK+_HDMI		

8 CN602 (to HDMI2)			
1	HDMI2_RX2+_HDMI	11	GND
2	GND	12	HDMI2_RX_CLK-_HDMI
3	HDMI2_RX2-_HDMI	13	CEC
4	HDMI2_RX1+_HDMI	14	ARC_SINGLE
5	GND	15	HDMI2_SCL_DDC
6	HDMI2_RX1-_HDMI	16	HDMI2_SDA_DDC
7	HDMI2_RX0+_HDMI	17	HDMI2_CABLE_DET
8	GND	18	HDMI2_5V_PW
9	HDMI2_RX0-_HDMI	19	HDMI2_HOT_PLUG
10	HDMI2_RX_CLK+_HDMI		

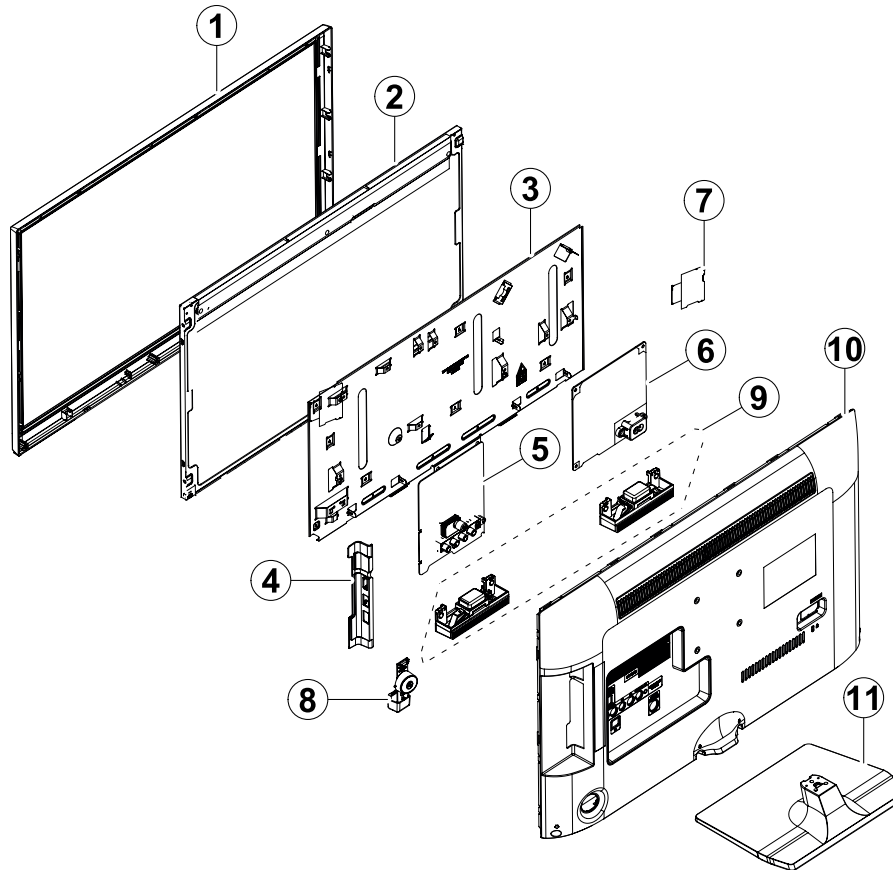
9 CN1403_LAN			
1	LAN_TXD+_LAN	5	LAN_GND
2	LAN_GND	6	LAN_RXD-_LAN
3	LAN_TXD-_LAN	7	NC
4	LAN_RXD+_LAN	8	LAN_GND

10 CN501_COMP			
1	Y(CVBS)	4	L
2	COMP_PB	5	R
3	COMP_PR		

11 CN403 (to Optical Jack)			
1	SPDIF_OUT	3	GND
2	B5V_DC_PW		

ANNEX. Exploded View & Part List [UN24M4500AFXZA WA01]

1-1. Exploded View



1-1-1. Parts List

No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
1	BN96-43844A	ASSY COVER P-FRONT; 24M4500,ABS,HB,BK0007	1	SA	
2	BN07-01378B	LCD-PANEL; V236BJ1-LE2,CMN61E2G,544.80x32	1	SA	
3	BN96-44172A	ASSY BRACKET P-PCB; 24M4500,EGI-SECC,NATU	1	SNA	
4	BN63-15392E	COVER-TERMINAL SIDE; 40M5000,HIPS,T1.5,HB	1	SA	
5	BN94-12232A	ASSY PCB MAIN; UM4*	1	SA	
6	BN44-00746F	DC VSS-PD BOARD; L23S0DSR_MPN,AC/DC,40.4W	1	SA	
7	BN59-01174D	NETWORK-WLAN CLIENT; WIDT30Q,55x24x5.8mm	1	SA	
8	BN96-30857U	ASSY BOARD P-FUNCTION TACT; M4500,SJ17-01	1	SNA	
9	BN96-30339H	ASSY SPEAKER P-FRONT; TV-SPK,HD470,12ohm,	1	SA	
10	BN96-43845A	ASSY COVER P-REAR; 24M4500,HIPS,V-0,BK002	1	SA	
11	BN96-31786A	ASSY STAND P-BOTTOM; TD310 23.6",ABS,BK0	1	SA	

2-1. Electrical Parts List

Service Bom (SA: SERVICE AVAILABLE, SNA: SERVICE NOT AVAILABLE)

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
			UN24M4500AFXZA (WA01)			
1	S001A	BN90-05991A	ASSY STAND;UH4100,23.6	1	SNA	
0.2	SB04A	BN96-31786A	ASSY STAND P-BOTTOM;TD310 23.6",ABS,BK0	1	SA	
..3	AS080	BN63-04755K	SHEET-PROTECTION COVER;19LC350/450,PE,T0	1	SNA	
..3		BN63-12195A	COVER-STAND TOP;24TD310,ABS,HB,BK0007,23	1	SNA	
..3	RF01	BN67-00327D	FOOT-RUBBER;STAND,Si,GRAY,W13,L13,T2	5	SNA	
..3		BN68-05603A	LABEL-E PASS;ART PAPER 90G,W/W	1	SNA	
1	R001A	BN90-09164C	ASSY COVER REAR;UM4*	1	SNA	
0.2	R001A	BN96-43845A	ASSY COVER P-REAR;24M4500,HIPS,V-0,BK002	1	SA	
..3	CB19	BN61-08997A	BRACKET-VESA;EGI-SECC,T0.8,NATURAL	4	SNA	
...4		0101-003397	STEEL PLATE;EGI,SECC,T0.8,E20,AG	12	SNA	
..3	R001	BN63-17018A	COVER-REAR;24M4500,HIPS,V-0,BK0020	1	SNA	
...4		0103-004637	RESIN HIPS;VE-1897,K2901,BK0020,1.5 V-0,	620	SNA	
..3		BN64-03902H	INLAY-TERMINAL SIDE;49M5300,PET,T0.125,W	1	SNA	
1		BN90-09249A	ASSY COVER FRONT;UM4*	1	SNA	
0.2	F001A	BN96-43844A	ASSY COVER P-FRONT;24M4500,ABS,HB,BK0007	1	SA	
..3	SCREW	6003-000282	SCREW-TAPTYPE;BH,+,-,B,M3,L8,ZPC(BLK),SW	1	SA	
..3	AB258	BN61-07182A	BRACKET-PANEL;EGI-SECC,T1,NATURAL	1	SNA	
..3	AS080	BN63-05199V	SHEET-PROTECTION COVER;AMBER,PE,T0.08,W4	1	SNA	
..3		BN63-11218A	SHEET-PROTECTION COVER;AMBER,AL,T0.08,W3	1	SNA	
..3	F001	BN63-17017A	COVER-FRONT;24M4500,ABS,HB,BK0007,W/W	1	SNA	
...4		0103-004609	RESIN ABS;HF-0680U,K21294,BK0007,HB,High	335	SNA	
1		BN90-09250A	ASSY W/I;UM4*	1	SNA	
0.2		BN81-08159Z	A/S PART SET-ELEC W/I;LED TV ELEC spec-C	1	SNA	
0.2		BN81-15563C	A/S PART SET-MECH W/I;UM4500D,U24MD*	1	SNA	
1		BN91-18234A	ASSY LCD;BN07-01378B	1	SNA	
0.2	PANEL	BN07-01378B	LCD-PANEL;V236BJ1-LE2,CMN61E2G,544.80x32	1	SA	
1		BN91-19085C	ASSY SHIELD;UM4*	1	SNA	
0.2	EC13	BN39-01762N	LEAD CONNECTOR-SATA;UN24H4000,UL21016,6P	1	SA	
0.2	EC13	BN39-02218B	LEAD CONNECTOR-SUB ASSY;55KU6000, 60KU60	1	SA	
0.2		BN39-02275E	LEAD CONNECTOR-POWER;24M4500,UL21016,10P	1	SA	
0.2		BN44-00746F	DC VSS-PD BOARD;L23S0DSR_MPN,AC/DC,40.4W	1	SA	
0.2	WIFI	BN59-01174D	NETWORK-WLAN CLIENT;WIDT30Q,55x24x5.8mm	1	SA	
0.2		BN61-10073A	HOLDER-WIFI;32H4000,ABS,BK0007,HB	1	SNA	
..3		0103-004609	RESIN ABS;HF-0680U,K21294,BK0007,HB,High	6	SNA	
0.2		BN63-15392E	COVER-TERMINAL SIDE;40M5000,HIPS,T1.5,HB	1	SA	
..3		0103-004628	RESIN HIPS;HF-1690H,K2901,BK0020,1.5mm H	16	SNA	
0.2	SP01A	BN96-30339H	ASSY SPEAKER P-FRONT;TV-SPK,HD470,12ohm,	1	SA	
0.2		BN96-30857U	ASSY BOARD P-FUNCTION TACT;M4500,SJ17-01	1	SNA	
0.2		BN96-34321U	FFC CABLE;kant-s,24M4500,Fold,L300,30P	1	SA	
0.2	CB07A	BN96-44172A	ASSY BRACKET P-PCB;24M4500,EGI-SECC,NATU	1	SNA	
..3		BN60-00162E	SPACER-FOAM;PE FOAM,L50M,GRAY,T0.5,W20	1	SNA	
..3	CB15	BN61-15020A	BRACKET-MAIN;24M4500,EGI-SECC,T0.6,NATUR	1	SNA	
1	M0017	BN91-19087C	ASSY CHASSIS;UM4*	1	SNA	

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
0.2	M0014	BN94-12232A	ASSY PCB MAIN;UM4*	1	SA	
..3		BN62-00835A	HEAT SINK-PS;M5300/M5500/M6300,A1050,T1.	1	SNA	
..3		BN68-05458A	LABEL-BARCODE;ALL,ART PAPER,W45,L12,BLAC	1	SNA	
..3		BN97-13207A	ASSY SMD;UM4*	1	SNA	
...4		0202-001830	SOLDER-CREAM;LFM-48W TM-HP,D20~38um,96.5	4	SNA	
...4	DS01A	0401-001056	DIODE-SWITCHING;MMBD4148SE,100V,200mA,SO	1	SA	
...4	DS01A	0401-001116	DIODE-SWITCHING;BAV99DW,75V,215mA,SOT-36	2	SA	
...4		0403-001785	DIODE-ZENER;NZH5V1B,4.94~5.2V,500mW,SOD-	1	SA	
...4		0404-001404	DIODE-SCHOTTKY;BAT721C,40V,200mA,SOT-23,	3	SA	
...4		0406-001200	DIODE-TVS;RClamp0504F,6V,1MAV,TP	2	SA	
...4		0406-001628	DIODE-TVS;AOZ8804ADI,6V,1MAV,5A,TP	1	SA	
...4		0406-001635	DIODE-TVS;SMF5.0A,6.4V,6.7V,7V,200MAV,20	1	SA	
...4		0406-001778	DIODE-TVS;PUSB3FR4,6V,9V,0.2MAV,0.7VPA,0	4	SA	
...4		0501-000445	TR-SMALL SIGNAL;KTC3875S-Y,NPN,150mW,SOT	3	SA	
...4		0504-000126	TR-DIGITAL;KSR1101,NPN,200mW,4.7K/4.7K,S	1	SA	
...4		0505-002560	FET-SILICON;AO6415,P,20V,-3.3A,0.15ohm,1	1	SA	
...4		0505-002598	FET-SILICON;AP2317GN,P,20V,-4.2A,0.052oh	3	SA	
...4		0505-003397	FET-SILICON;2N7002K,N,60V,0.38A,1.19ohm,	5	SA	
...4		1001-001545	IC-ANALOG MULTIPLEX;TS3USB221ERSE,R-PQF	1	SA	
...4		1103-001584	IC-EEPROM;M24512-DFMN6TP,512Kbit,64K x 8	1	SNA	
...4		1105-002681	IC-DDR3 SDRAM;H5TQ4G63CFR,DDR3 1866,4Gbi	1	SNA	
...4		1105-002874	IC-DDR3 SDRAM;H5TQ2G83GFR-TEC,DDR3 2133,	2	SNA	
...4		1201-003690	IC-AUDIO AMP;TAS5747PHPR,QFP,48P,DUAL,PL	1	SA	
...4		1203-004363	IC-VOL. DETECTOR;SOT-23,3Z30,2.9x1.6mm,P	1	SA	
...4		1203-006130	IC-POSI.FIXED REG.;S-1172B25-U5T1G,SOT-8	1	SA	
...4		1203-008104	IC-POSI.FIXED REG.;S-13A1D18-E800,HSOP,8	1	SNA	
...4		1203-008105	IC-POSI.FIXED REG.;S-13A1D33-E800,HSOP,8	1	SNA	
...4		1203-008392	IC-VOL. DETECTOR;S-6414AAB-L800X,TSOT-23	1	SA	
...4		1203-008393	IC-DDR TERMINATION REG.;AP2303MTR-G1-89,	1	SA	
...4		1203-008734	IC-DC/DC CONVERTER;RT6214,TSOT-23,6,3x1.	2	SA	
...4		1203-008919	IC-DC/DC CONVERTER;SYD104AADC,TSOT23,6P,	1	SA	
...4		1203-008921	IC-DC/DC CONVERTER;SYU236ARAC,QFN,20P,3x	1	SA	
...4		1204-003720	IC-DECODER;SDP1604,HS-LFBGA,576P,22.8x22	1	SA	
...4		1205-005519	IC-SWITCH;ET20163,SOT23-5,5P,2.95x3.02mm	1	SA	
...4		1405-001232	VARISTOR;6.4V,5.6VDC,30A,1608,TP,19V,200	4	SNA	
...4		1405-001271	VARISTOR;35V,20VDC,5A,1005,TP,100V,10pF	14	SA	
...4		1405-001381	VARISTOR;11V,8VDC,30A,1608,TP,25V,500pF	2	SA	
...4		1405-001382	VARISTOR;24.5V,16VDC,120A,2012,TP,42V,40	1	SA	
...4		2007-000039	R-CHIP;0ohm,1%,1/10W,TP,1608	4	SA	
...4		2007-000070	R-CHIP;0ohm,5%,1/10W,TP,1608	2	SA	
...4		2007-000137	R-CHIP;2Kohm,5%,1/16W,TP,1005	10	SA	
...4		2007-000138	R-CHIP;100ohm,5%,1/16W,TP,1005	21	SA	
...4		2007-000143	R-CHIP;4.7Kohm,5%,1/16W,TP,1005	47	SNA	
...4		2007-000148	R-CHIP;10Kohm,5%,1/16W,TP,1005	26	SA	
...4		2007-000171	R-CHIP;0ohm,5%,1/16W,TP,1005	26	SNA	
...4		2007-000172	R-CHIP;10ohm,5%,1/16W,TP,1005	2	SA	
...4		2007-000173	R-CHIP;22ohm,5%,1/16W,TP,1005	13	SA	

ANNEX. Exploded View & Part List

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
...4		2007-000475	R-CHIP;1Mohm,1%,1/10W,TP,1608	1	SA	
...4		2007-000932	R-CHIP;470ohm,5%,1/16W,TP,1005	5	SA	
...4		2007-001288	R-CHIP;18ohm,5%,1/16W,TP,1005	4	SA	
...4		2007-001292	R-CHIP;33ohm,5%,1/16W,TP,1005	12	SA	
...4		2007-001298	R-CHIP;51ohm,5%,1/16W,TP,1005	1	SA	
...4		2007-003022	R-CHIP;62ohm,5%,1/16W,TP,1005	25	SNA	
...4		2007-007107	R-CHIP;100Kohm,1%,1/16W,TP,1005	8	SA	
...4		2007-007134	R-CHIP;39Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007135	R-CHIP;18Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-007137	R-CHIP;1.2Kohm,1%,1/16W,TP,1005	2	SA	
...4		2007-007138	R-CHIP;27Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007139	R-CHIP;47Kohm,1%,1/16W,TP,1005,T0.35	5	SA	
...4		2007-007142	R-CHIP;10Kohm,1%,1/16W,TP,1005	5	SA	
...4		2007-007309	R-CHIP;12Kohm,1%,1/16W,TP,1005,T0.35	6	SA	
...4		2007-007311	R-CHIP;22Kohm,1%,1/16W,TP,1005	7	SA	
...4		2007-007312	R-CHIP;20Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007316	R-CHIP;3.3Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-007318	R-CHIP;1Kohm,1%,1/16W,TP,1005	28	SNA	
...4		2007-007443	R-CHIP;3.01Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-007470	R-CHIP;7.5Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007517	R-CHIP;240ohm,1%,1/16W,TP,1005	5	SNA	
...4		2007-007526	R-CHIP;23.7Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-007528	R-CHIP;1.5Kohm,1%,1/16W,TP,1005	2	SA	
...4		2007-007538	R-CHIP;56Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-008015	R-CHIP;75ohm,1%,1/16W,TP,1005	3	SNA	
...4		2007-008137	R-CHIP;24Kohm,1%,1/16W,TP,1005	1	SNA	
...4	AD480	2203-000233	C-CER,CHIP;0.1nF,5%,50V,C0G,TP,1005	4	SNA	
...4	AD480	2203-000278	C-CER,CHIP;0.01nF,0.5pF,50V,C0G,TP,1005	4	SA	
...4	AD480	2203-000359	C-CER,CHIP;0.15nF,5%,50V,C0G,TP,1005,T0.	3	SA	
...4	AD480	2203-000438	C-CER,CHIP;1nF,10%,50V,X7R,TP,1005	9	SA	
...4	AD480	2203-000489	C-CER,CHIP;2.2nF,10%,50V,X7R,TP,1005	1	SA	
...4	AD480	2203-000585	C-CER,CHIP;0.22nF,10%,50V,X7R,TP,1005	2	SA	
...4	AD480	2203-000627	C-CER,CHIP;0.022nF,5%,50V,C0G,TP,1005	3	SA	
...4	AD480	2203-000679	C-CER,CHIP;0.027nF,5%,50V,C0G,TP,1005	2	SA	
...4	AD480	2203-000714	C-CER,CHIP;3.3nF,10%,50V,X7R,TP,1005	2	SA	
...4	AD480	2203-000812	C-CER,CHIP;0.033nF,5%,50V,C0G,TP,1005	1	SA	
...4	AD480	2203-000940	C-CER,CHIP;0.47nF,10%,50V,X7R,TP,1005	4	SNA	
...4	AD480	2203-000995	C-CER,CHIP;0.047nF,5%,50V,C0G,TP,1005	2	SNA	
...4	AD480	2203-001412	C-CER,CHIP;0.03nF,5%,50V,C0G,TP,1005	3	SNA	
...4	AD480	2203-002285	C-CER,CHIP;10nF,10%,50V,X7R,TP,1005	11	SNA	
...4	AD480	2203-005083	C-CER,CHIP;220nF,10%,50V,X7R,TP,1608,T0.	6	SNA	
...4	AD480	2203-005968	C-CER,CHIP;4.7nF,10%,50V,X7R,TP,1005,T0.	2	SNA	
...4	AD480	2203-006048	C-CER,CHIP;100nF,10%,10V,X7R,TP,1005	131	SNA	
...4	AD480	2203-006126	C-CER,CHIP;47nF,10%,16V,X7R,TP,1005	11	SNA	
...4	AD480	2203-006324	C-CER,CHIP;2200nF,10%,10V,X5R,TP,1608	2	SA	
...4	AD480	2203-006348	C-CER,CHIP;1000nF,10%,25V,X5R,TP,1608,T0	4	SA	
...4	AD480	2203-006562	C-CER,CHIP;1000nF,10%,10V,X5R,TP,1005	7	SA	

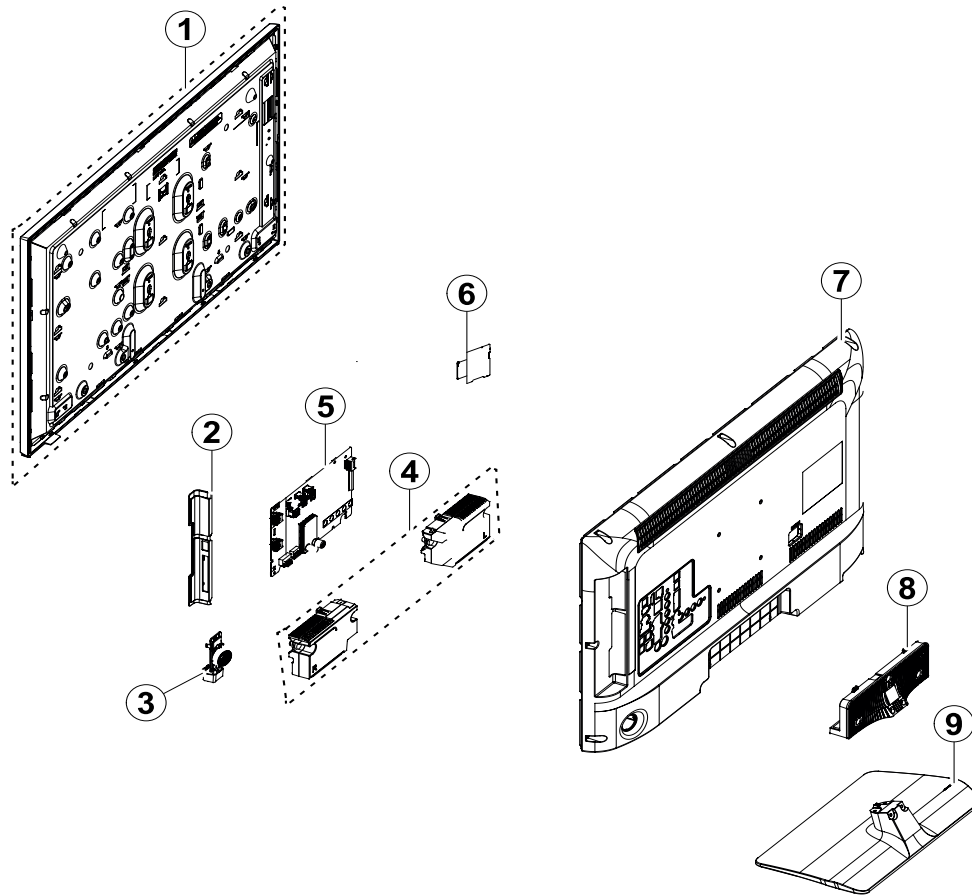
Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
...4	AD480	2203-006824	C-CER,CHIP;4700nF,10%,10V,X5R,TP,1608	4	SA	
...4	AD480	2203-006838	C-CER,CHIP;2200nF,10%,6.3V,X5R,TP,1005	12	SA	
...4	AD480	2203-006890	C-CER,CHIP;10000nF,20%,6.3V,X5R,TP,1608	32	SNA	
...4	AD480	2203-007240	C-CER,CHIP;22000nF,20%,6.3V,X5R,TP,1810,	12	SA	
...4	AD480	2203-007269	C-CER,CHIP;22000nF,20%,10V,X5R,TP,2012(2	9	SA	
...4	AD480	2203-007270	C-CER,CHIP;10000nF,10%,10V,X5R,TP,1608,T	4	SNA	
...4	AD480	2203-007306	C-CER,CHIP;10000nF,10%,25V,X5R,TP,2012,1	7	SNA	
...4	AD480	2203-007544	C-CER,CHIP;100nF,10%,50V,X7R,TP,1005,T0.	11	SA	
...4	AD480	2203-008096	C-CER,CHIP;2200nF,10%,50V,X5R,TP,2012,1.	1	SA	
...4	AD480	2203-008315	C-CER,CHIP;22000nF,20%,25V,X5R,TP,2012,T	2	SA	
...4		2402-001268	C-AL,SMD;100uF,20%,25V,WT,TP,8x6.3mm	1	SA	
...4		2703-000213	INDUCTOR-SMD;470nH,10%,1.35Ohm,35mA,15,M	1	SA	
...4		2703-002269	INDUCTOR-SMD;56nH,5%,1005,T0.5,1.4Ohm,15	2	SA	
...4		2703-003149	INDUCTOR-SMD;2.2uH,20%,0.055Ohm,3000mA,W	1	SA	
...4		2703-004724	INDUCTOR-SMD;8.2uH,20%,5050,T4,0.072Ohm,	4	SA	
...4		2703-005191	INDUCTOR-SMD;1.5uH,20%,6060,T4.5,0.02Ohm	1	SA	
...4		2703-005192	INDUCTOR-SMD;4.7uH,20%,6060,T4.5,0.036Oh	2	SA	
...4		2801-003773	CRYSTAL-SMD;12MHz,30ppm,28-AAN,20pF,50Oh	1	SA	
...4		3301-001364	BEAD-SMD;1000ohm,1608,TP,1085ohm/108MHz,	6	SNA	
...4		3301-002039	BEAD-SMD;26ohm,1608,TP	30	SA	
...4		3404-001379	SWITCH-TACT;12V,50mA,250gf,4.9x4.9x1.5mm	1	SA	
...4		3601-001374	FUSE-SURFACE MOUNT;32V,5A,FAST-ACTING,PL	2	SA	
...4		3701-001967	CONNECTOR-HDMI;19P,A,FEMALE,AU,0.5mm,BLK	2	SA	
...4		3707-001106	CONNECTOR-OPTICAL;STRAIGHT,SPDIF,2.5PI	1	SA	
...4		3708-001150	CONNECTOR-FPC/FFC/PIC;30P,1mm,SMD-A,SN,Z	1	SA	
...4		3711-007776	CONNECTOR-HEADER;BOX,8P,1R,1.25mm,SMD-A,	1	SA	
...4	EH01	3711-008131	HEADER-BOARD TO CABLE;BOX,4P,1R,2.5mm,AN	1	SA	
...4	EH01	3711-008634	HEADER-BOARD TO CABLE;BOX,10P,2R,2mm,ANG	1	SA	
...4		3711-009090	CONNECTOR-HEADER;BOX,16P,1R,1.25mm,SMD-A	1	SA	
...4		3722-003229	JACK-MODULAR;8P/8C,Y,STRAIGHT,NONE,AU,1P	1	SA	
...4		3722-003457	JACK-USB;4P/1C,NI,BLK,ANGLE,A,2.0,13.1x1	1	SA	
...4	JACK PIN	3722-003546	JACK-PIN;5P,NI/SN,BLU/GRN/RED/WHT/YEL,SM	1	SA	
...4		6302-001376	GASKET-SMD;;,CONDUCTIVE FABRIC,T14,W43,L	6	SNA	
...4		BN40-00329A	TUNER-DTV AIR CABLE;DTVS20EH7A,VSB Half,	1	SA	
...4		BN41-02574A	PCB-MAIN;M5300,FR-4,4L,1.2T,1,Kant-S Pla	1	SNA	
...4	CB07	BN61-13312B	BRACKET-SCREWLESS PCB;55KS8000,SK5,T0.3,	4	SNA	
...4		BN97-13036A	ASSY MICOM-MAIN;T-KTSAKUC-0609.10,1107-0	1	SNA	
...5		1107-002505	IC-EMMC;THGBMDG5D1LBAIT,4Gbyte,x1,WFBGA,	1	SA	
...4		BN97-13274A	ASSY MICOM-SUB;T-KTSINTV,UNV72,1107-0022	1	SNA	
...5		1107-002226	IC-NOR FLASH;W25Q40CLSSIP,4Mbit,SOIC,8P,	1	SA	
..3		BN97-13396A	ASSY DRM;KANT-S,ATSC,MAC,HDMP	1	SNA	
...4		BN46-00109H	KEY CODE-CERTIFICATION;MAC,TV/AV,General	1	SNA	
...4		BN46-00110P	KEY CODE-CERTIFICATION;MIRACAST(HDCP2.2)	1	SNA	
1		BN92-22112C	ASSY P/MATERIAL;UM4*	1	SNA	
0.2		6902-002592	BAG ROLL;HDPE/PE FOAM,T0.015/T1,W700,L60	1	SNA	
0.2		6922-000013	BAND;PP,W18,L2300/L2900,TRP	1	SNA	
0.2		BN02-00319B	TAPE-SINGLE FACE;OPP,T0.05,W75,L800M,CLE	2	SNA	

ANNEX. Exploded View & Part List

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
0.2	T0603	BN69-01126E	PALLET-WOODEN;MGM23,WOOD,W1090,L760,H120	1	SNA	
0.2		BN69-16098A	CUSHION-SET;24M4500,EPS,16.7g/l,WHITE	1	SNA	
..3		BN81-01918A	A/S-RESIN;EPS,SG-302	0	SNA	
0.2		BN69-16844A	PAD SHEET-PE;24H4000,CB,1070,750,YELLOW,	1	SNA	
1		BN92-22113F	ASSY BOX;UM4*	1	SNA	
0.2		BN68-07893A	LABEL BOX;ALL,THERMAL PAPER,W60,L110,WHT	1	SNA	
0.2		BN69-16253A	BOX UNIT;24M4500,CB,SW4,A1,L751,W112,H38	1	SNA	
1		BN92-22114F	ASSY LABEL;UM4*	1	SNA	
0.2		BN02-00102B	TAPE-SINGLE FACE;FILAMENT,#8917,T0.15,W2	0	SNA	
0.2		BN68-07105F	LABEL-RATING;Monitor,WW,PP,T0.161,W90,L6	1	SNA	
0.2		BN68-08562E	LEAFLET-QUICK SETUP GUIDE;UN49J5200AFXZX	1	SNA	
0.2		BN68-08653A	LABEL-ENERGY;UN24M4500AFXZA,UNITED STATE	1	SNA	
1	ACCE1	BN92-22115F	ASSY ACCESSORY;UM4*	1	SNA	
0.2		BN96-43806B	ASSY ACCESSORY-MANUAL CABLE;UM4* 24,ZC	1	SNA	
..3	T0268	3903-001117	CBF-POWER CORD;DT,USA,Angle,2P(C7),125V,	1	SA	
..3		4301-000121	BATTERY-MN;1.5V,R03,10.5x44.5m,7.5g,AAA	2	SNA	
..3		6902-001962	BAG PE;LDPE,BIOBASED,T0.05,W400,L300,TRP	1	SNA	
..3		BN59-01267A	REMOCON-TV;2017 TV,Samsung,45KEY,3V,Nort	1	SA	
..3	T0527	BN68-00513A	LABEL-E PASS;ALL MODEL,WW,YUPO,W50,L15,W	1	SNA	
..3		BN68-04972E	LEAFLET-REGULATORY GUIDE;ALL,SAMSUNG,W/W	1	SNA	
..3		BN68-08058F	MANUAL USERS;UN49J5000AFXZA,UNITED STATE	1	SNA	
..3		BN96-18013D	ASSY ACCESSORY-SCREW;ALL MODEL,6003-0017	1	SNA	
...4	SCREW	6003-001782	SCREW-TAPTYPE;BH,+,B,M4,L12,ZPC(BLK),SWR	3	SA	
...4		BN69-08203D	BAG SCREW;LDPE,0.05,70,90,TRP,Vinyl pack	1	SNA	

ANNEX. Exploded View & Part List [UN28M4500AFXZA BA01]

1-1. Exploded View



1-1-1. Parts List

No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
1	BN95-04265A	PRODUCT LCD-BOE; CY-HM028AGEV1V/H,M4500,B	1	SA	
2	BN63-15392E	COVER-TERMINAL SIDE; 40M5000,HIPS,T1.5,HB	1	SA	
3	BN96-30857U	ASSY BOARD P-FUNCTION TACT; M4500,SJ17-01	1	SNA	
4	BN96-30338A	ASSY SPEAKER P-FRONT; TV-SPK,H4000,12ohm,	1	SA	
5	BN94-12232B	ASSY PCB MAIN; UM4*	1	SA	
6	BN59-01174D	NETWORK-WLAN CLIENT; WIDT30Q,55x24x5.8mm	1	SA	
7	BN96-43863A	ASSY COVER P-REAR; 28M4500,HIPS,V-0,BK002	1	SA	
8	BN96-43918A	ASSY GUIDE P-STAND; 28M4500,PC+GF20%,BK00	1	SA	
9	BN96-31474A	ASSY STAND P-BOTTOM; UH4K 32,ABS,BLACK,Sq	1	SA	

2-1. Electrical Parts List

Service Bom (SA: SERVICE AVAILABLE, SNA: SERVICE NOT AVAILABLE)

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
			UN28M4500AFXZA (BA01)			
1	R001A	BN90-09164B	ASSY COVER REAR;UM4*	1	SNA	
0.2	R001A	BN96-43863A	ASSY COVER P-REAR;28M4500,HIPS,V-0,BK002	1	SA	
..3	R001	BN63-17019A	COVER-REAR;28M4500,HIPS,V-0,BK0020	1	SNA	
...4		0103-004637	RESIN HIPS;VE-1897,K2901,BK0020,1.5 V-0,	2394	SNA	
..3		BN64-03902H	INLAY-TERMINAL SIDE;49M5300,PET,T0.125,W	1	SNA	
1	S001A	BN90-09165A	ASSY STAND;UM4*	1	SNA	
0.2	SB04A	BN96-31474A	ASSY STAND P-BOTTOM;UH4K 32,ABS,BLACK,Sq	1	SA	
..3	AS080	BN63-04755L	SHEET-PROTECTION COVER;22LC350,PE,T0.05,	0	SNA	
..3		BN63-11623A	COVER-STAND TOP;32UH4000,ABS,Non-EU,HB,B	1	SNA	
...4		0103-004609	RESIN ABS;HF-0680U,K21294,BK0007,HB,High	256	SNA	
..3	RF01	BN67-00327D	FOOT-RUBBER;STAND,Si,GRAY,W13,L13,T2	5	SNA	
..3		BN68-05603A	LABEL-E PASS;ART PAPER 90G,W/W	2	SNA	
0.2	SG01A	BN96-43918A	ASSY GUIDE P-STAND;28M4500,PC+GF20%,BK00	1	SA	
..3	SG03	BN61-15081A	GUIDE-STAND;28M4500,PC+GF20%,T3.5,V-2,BK	1	SNA	
...4		0103-007285	RESIN PC;HF3200H,K21294,BK0007,V-2,GF20%	294	SNA	
..3		BN69-08751A	PACKING SHRINKAGE;STAND,PLASTIC OTHERS,P	1	SNA	
..3		BN96-29120U	ASSY ACCESSORY-SCREW;LED_J4000,6003-0017	1	SNA	
...4	SCREW	6003-001782	SCREW-TAPTYPE;BH,+,B,M4,L12,ZPC(BLK),SWR	6	SA	
...4		6902-002582	BAG SCREW;LDPE,T0.05*,W70,90,TRP,RECYCLE	1	SNA	
..3		BN96-37289B	ASSY ACCESSORY-SCREW;28M4500,BN61-09494B	1	SNA	
...4		6902-002837	BAG SCREW;LDPE,T0.05,W70,L90,TRP,RECYCLE	1	SNA	
...4	SCREW	BN61-09494B	SCREW-TAPTYPE;BH,+,S,M4,L14,ZPC(BLK),SWR	2	SA	
1		BN90-09236A	ASSY W/I;UM4*	1	SNA	
0.2		BN81-08159Z	A/S PART SET-ELEC W/I;LED TV ELEC spec-C	1	SNA	
0.2		BN81-15563D	A/S PART SET-MECH W/I;UM4500D,U28MD*	1	SNA	
1		BN91-19085B	ASSY SHIELD;UM4*	1	SNA	
0.2	EC13	BN39-02218B	LEAD CONNECTOR-SUB ASSY;55KU6000, 60KU60	1	SA	
0.2		BN39-02358A	LEAD CONNECTOR-POWER;UL21016,8P,500mm,BL	1	SA	
0.2	WIFI	BN59-01174D	NETWORK-WLAN CLIENT;WIDT30Q,55x24x5.8mm	1	SA	
0.2		BN61-11586A	HOLDER-WIFI;JU7500,ABS,BK0007,HB	1	SNA	
0.2		BN63-15392E	COVER-TERMINAL SIDE;40M5000,HIPS,T1.5,HB	1	SA	
..3		0103-004628	RESIN HIPS;HF-1690H,K2901,BK0020,1.5mm H	16	SNA	
0.2	SP01A	BN96-30338A	ASSY SPEAKER P-FRONT;TV-SPK,H4000,12ohm,	1	SA	
0.2		BN96-30857U	ASSY BOARD P-FUNCTION TACT;M4500,SJ17-01	1	SNA	
0.2		BN96-35954H	FFC CABLE;Kant-s,28M4500,Fold,L350,30P	1	SA	
1	M0017	BN91-19087B	ASSY CHASSIS;UM4*	1	SNA	
0.2	M0014	BN94-12232B	ASSY PCB MAIN;UM4*	1	SA	
..3		BN62-00835A	HEAT SINK-PS;M5300/M5500/M6300,A1050,T1.	1	SNA	
..3		BN68-05458A	LABEL-BARCODE;ALL,ART PAPER,W45,L12,BLAC	1	SNA	
..3		BN97-13207B	ASSY SMD;UM4*	1	SNA	
...4		0202-001830	SOLDER-CREAM;LFM-48W TM-HP,D20~38um,96.5	4	SNA	
...4	DS01A	0401-001056	DIODE-SWITCHING;MMBD4148SE,100V,200mA,SO	1	SA	
...4	DS01A	0401-001099	DIODE-SWITCHING;1N4148WS,75V,150mA,SOD-3	1	SA	

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
...4	DS01A	0401-001116	DIODE-SWITCHING;BAV99DW,75V,215mA,SOT-36	2	SA	
...4	DS01A	0401-001166	DIODE-SWITCHING;BAV20WS-V,150V,250mA,SOD	2	SA	
...4	DR01A	0402-002006	DIODE-RECTIFIER;SBR6100CTL,100V,6A,TO-25	1	SA	
...4		0403-001411	DIODE-ZENER;BZT52-B5V6S,5.49~5.73V,200mW	1	SA	
...4		0403-001785	DIODE-ZENER;NZH5V1B,4.94~5.2V,500mW,SOD-	1	SA	
...4		0404-001307	DIODE-SCHOTTKY;SSC54,40V,5000mA,DO-214AB	1	SA	
...4		0404-001404	DIODE-SCHOTTKY;BAT721C,40V,200mA,SOT-23,	3	SA	
...4		0406-001200	DIODE-TVS;RClamp0504F,6V,1MAV,TP	2	SA	
...4		0406-001628	DIODE-TVS;AOZ8804ADI,6V,1MAV,5A,TP	1	SA	
...4		0406-001635	DIODE-TVS;SMF5.0A,6.4V,6.7V,7V,200MAV,20	1	SA	
...4		0406-001778	DIODE-TVS;PUSB3FR4,6V,9V,0.2MAV,0.7VPA,0	4	SA	
...4		0501-000445	TR-SMALL SIGNAL;KTC3875S-Y,NPN,150mW,SOT	4	SA	
...4		0504-000126	TR-DIGITAL;KSR1101,NPN,200mW,4.7K/4.7K,S	1	SA	
...4		0505-002560	FET-SILICON;AO6415,P,20V,-3.3A,0.15ohm,1	1	SA	
...4		0505-002598	FET-SILICON;AP2317GN,P,20V,-4.2A,0.052oh	3	SA	
...4		0505-003224	FET-SILICON;AO6405,P,30V,-5A,0.052ohm,2W	1	SA	
...4		0505-003397	FET-SILICON;2N7002K,N,60V,0.38A,1.19ohm,	7	SA	
...4		0505-003620	FET-SILICON;AOD4286,N,100V,14A,55.5mohm,	1	SA	
...4		1001-001545	IC-ANALOG MULTIPLEX;TS3USB221ERSE,R-PQF	1	SA	
...4		1103-001584	IC-EEPROM;M24512-DFMN6TP,512Kbit,64K x 8	1	SNA	
...4		1105-002681	IC-DDR3 SDRAM;H5TQ4G63CFR,DDR3 1866,4Gbi	1	SNA	
...4		1105-002874	IC-DDR3 SDRAM;H5TQ2G83GFR-TEC,DDR3 2133,	2	SNA	
...4		1201-003690	IC-AUDIO AMP;TAS5747PHPR,QFP,48P,DUAL,PL	1	SA	
...4		1203-004363	IC-VOL. DETECTOR;SOT-23,3Z30,2.9x1.6mm,P	1	SA	
...4		1203-006130	IC-POS.FIXED REG.;S-1172B25-U5T1G,SOT-8	1	SA	
...4		1203-008030	IC-DC/DC CONVERTER;TPS54531DDA,SO PowerP	1	SA	
...4		1203-008104	IC-POS.FIXED REG.;S-13A1D18-E800,HSOP,8	1	SNA	
...4		1203-008105	IC-POS.FIXED REG.;S-13A1D33-E800,HSOP,8	1	SNA	
...4		1203-008392	IC-VOL. DETECTOR;S-6414AAB-L800X,TSOT-23	1	SA	
...4		1203-008393	IC-DDR TERMINATION REG.;AP2303MTR-G1-89,	1	SA	
...4		1203-008728	IC-BACKLIGHT DRIVER;BD9413F,SOP,18P,11.2	1	SA	
...4		1203-008734	IC-DC/DC CONVERTER;RT6214,TSOT-23,6,3x1.	2	SA	
...4		1203-008919	IC-DC/DC CONVERTER;SYD104AADC,TSOT23,6P,	1	SA	
...4		1203-008921	IC-DC/DC CONVERTER;SYU236ARAC,QFN,20P,3x	1	SA	
...4		1204-003720	IC-DECODER;SDP1604,HS-LFBGA,576P,22.8x22	1	SA	
...4		1205-005519	IC-SWITCH;ET20163,SOT23-5,5P,2.95x3.02mm	1	SA	
...4		1405-001232	VARISTOR;6.4V,5.6VDC,30A,1608,TP,19V,200	4	SNA	
...4		1405-001271	VARISTOR;35V,20VDC,5A,1005,TP,100V,10pF	14	SA	
...4		1405-001381	VARISTOR;11V,8VDC,30A,1608,TP,25V,500pF	2	SA	
...4		2007-000039	R-CHIP;0ohm,1%,1/10W,TP,1608	4	SA	
...4		2007-000060	R-CHIP;100Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-000070	R-CHIP;0ohm,5%,1/10W,TP,1608	2	SA	
...4		2007-000137	R-CHIP;2Kohm,5%,1/16W,TP,1005	10	SA	
...4		2007-000138	R-CHIP;100ohm,5%,1/16W,TP,1005	22	SA	
...4		2007-000143	R-CHIP;4.7Kohm,5%,1/16W,TP,1005	47	SNA	
...4		2007-000148	R-CHIP;10Kohm,5%,1/16W,TP,1005	27	SA	
...4		2007-000153	R-CHIP;22Kohm,5%,1/16W,TP,1005	3	SNA	

ANNEX. Exploded View & Part List

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
...4		2007-000171	R-CHIP;0ohm,5%,1/16W,TP,1005	27	SNA	
...4		2007-000172	R-CHIP;10ohm,5%,1/16W,TP,1005	2	SA	
...4		2007-000173	R-CHIP;22ohm,5%,1/16W,TP,1005	13	SA	
...4		2007-000309	R-CHIP;10ohm,5%,1/10W,TP,1608	1	SNA	
...4		2007-000343	R-CHIP;120ohm,1%,1/10W,TP,1608	1	SA	
...4		2007-000475	R-CHIP;1Mohm,1%,1/10W,TP,1608	1	SA	
...4		2007-000614	R-CHIP;24Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-000879	R-CHIP;4.7ohm,1%,1/10W,TP,1608	1	SA	
...4		2007-000932	R-CHIP;470ohm,5%,1/16W,TP,1005	4	SA	
...4		2007-001288	R-CHIP;18ohm,5%,1/16W,TP,1005	4	SA	
...4		2007-001292	R-CHIP;33ohm,5%,1/16W,TP,1005	11	SA	
...4		2007-001298	R-CHIP;51ohm,5%,1/16W,TP,1005	1	SA	
...4		2007-001323	R-CHIP;3Kohm,5%,1/16W,TP,1005	1	SA	
...4		2007-002749	R-CHIP;3.3ohm,5%,1/4W,TP,3216	2	SNA	
...4		2007-002906	R-CHIP;200Kohm,1%,1/10W,TP,1608	2	SA	
...4		2007-003022	R-CHIP;62ohm,5%,1/16W,TP,1005	25	SNA	
...4		2007-007107	R-CHIP;100Kohm,1%,1/16W,TP,1005	20	SA	
...4		2007-007134	R-CHIP;39Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007135	R-CHIP;18Kohm,1%,1/16W,TP,1005	3	SNA	
...4		2007-007137	R-CHIP;1.2Kohm,1%,1/16W,TP,1005	2	SA	
...4		2007-007138	R-CHIP;27Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007139	R-CHIP;47Kohm,1%,1/16W,TP,1005,T0.35	6	SA	
...4		2007-007142	R-CHIP;10Kohm,1%,1/16W,TP,1005	6	SA	
...4		2007-007309	R-CHIP;12Kohm,1%,1/16W,TP,1005,T0.35	6	SA	
...4		2007-007311	R-CHIP;22Kohm,1%,1/16W,TP,1005	8	SA	
...4		2007-007312	R-CHIP;20Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007316	R-CHIP;3.3Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-007318	R-CHIP;1Kohm,1%,1/16W,TP,1005	30	SNA	
...4		2007-007334	R-CHIP;200Kohm,1%,1/16W,TP,1005	7	SA	
...4		2007-007382	R-CHIP;20Mohm,5%,1/10W,TP,1608	1	SA	
...4		2007-007405	R-CHIP;560ohm,1%,1/16W,TP,1005	2	SA	
...4		2007-007443	R-CHIP;3.01Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-007470	R-CHIP;7.5Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-007517	R-CHIP;240ohm,1%,1/16W,TP,1005	5	SNA	
...4		2007-007526	R-CHIP;23.7Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-007528	R-CHIP;1.5Kohm,1%,1/16W,TP,1005	2	SA	
...4		2007-007538	R-CHIP;56Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-008015	R-CHIP;75ohm,1%,1/16W,TP,1005	3	SNA	
...4		2007-008035	R-CHIP;160Kohm,1%,1/10W,TP,1608	1	SA	
...4		2007-008136	R-CHIP;36Kohm,1%,1/16W,TP,1005	2	SA	
...4		2007-008137	R-CHIP;24Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-008167	R-CHIP;120Kohm,1%,1/16W,TP,1005	2	SC	
...4		2007-008263	R-CHIP;3Kohm,1%,1/16W,TP,1005	1	SNA	
...4		2007-008269	R-CHIP;51Kohm,1%,1/16W,TP,1005	2	SNA	
...4		2007-008517	R-CHIP;240Kohm,1%,1/16W,TP,1005	1	SA	
...4		2007-008720	R-CHIP;4.7ohm,1%,1/4W,TP,3216	2	SNA	
...4		2007-008779	R-CHIP;0ohm,1%,1/16W,TP,1005	1	SA	

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
...4		2007-009234	R-CHIP;0.47ohm,1%,1/4W,TP,3216	4	SNA	
...4	AD480	2203-000138	C-CER,CHIP;1.5nF,10%,50V,X7R,TP,1005	1	SA	
...4	AD480	2203-000233	C-CER,CHIP;0.1nF,5%,50V,C0G,TP,1005	4	SNA	
...4	AD480	2203-000278	C-CER,CHIP;0.01nF,0.5pF,50V,C0G,TP,1005	4	SA	
...4	AD480	2203-000311	C-CER,CHIP;0.12nF,5%,50V,C0G,TP,1005	1	SA	
...4	AD480	2203-000359	C-CER,CHIP;0.15nF,5%,50V,C0G,TP,1005,T0.	3	SA	
...4	AD480	2203-000438	C-CER,CHIP;1nF,10%,50V,X7R,TP,1005	12	SA	
...4	AD480	2203-000489	C-CER,CHIP;2.2nF,10%,50V,X7R,TP,1005	1	SA	
...4	AD480	2203-000585	C-CER,CHIP;0.22nF,10%,50V,X7R,TP,1005	2	SA	
...4	AD480	2203-000627	C-CER,CHIP;0.022nF,5%,50V,C0G,TP,1005	3	SA	
...4	AD480	2203-000679	C-CER,CHIP;0.027nF,5%,50V,C0G,TP,1005	3	SA	
...4	AD480	2203-000714	C-CER,CHIP;3.3nF,10%,50V,X7R,TP,1005	2	SA	
...4	AD480	2203-000726	C-CER,CHIP;3.9nF,10%,50V,X7R,TP,1608	3	SA	
...4	AD480	2203-000812	C-CER,CHIP;0.033nF,5%,50V,C0G,TP,1005	1	SA	
...4	AD480	2203-000940	C-CER,CHIP;0.47nF,10%,50V,X7R,TP,1005	4	SNA	
...4	AD480	2203-000995	C-CER,CHIP;0.047nF,5%,50V,C0G,TP,1005	2	SNA	
...4	AD480	2203-001124	C-CER,CHIP;0.68nF,10%,50V,X7R,TP,1005	1	SA	
...4	AD480	2203-001412	C-CER,CHIP;0.03nF,5%,50V,C0G,TP,1005	3	SNA	
...4	AD480	2203-002285	C-CER,CHIP;10nF,10%,50V,X7R,TP,1005	12	SNA	
...4	AD480	2203-003019	C-CER,CHIP;8.2nF,10%,50V,X7R,TP,1005	1	SA	
...4	AD480	2203-005083	C-CER,CHIP;220nF,10%,50V,X7R,TP,1608,T0.	6	SNA	
...4	AD480	2203-005249	C-CER,CHIP;100nF,10%,50V,X7R,TP,1608	3	SA	
...4	AD480	2203-005968	C-CER,CHIP;4.7nF,10%,50V,X7R,TP,1005,T0.	2	SNA	
...4	AD480	2203-006048	C-CER,CHIP;100nF,10%,10V,X7R,TP,1005	131	SNA	
...4	AD480	2203-006126	C-CER,CHIP;47nF,10%,16V,X7R,TP,1005	12	SNA	
...4	AD480	2203-006324	C-CER,CHIP;2200nF,10%,10V,X5R,TP,1608	2	SA	
...4	AD480	2203-006348	C-CER,CHIP;1000nF,10%,25V,X5R,TP,1608,T0	6	SA	
...4	AD480	2203-006562	C-CER,CHIP;1000nF,10%,10V,X5R,TP,1005	7	SA	
...4	AD480	2203-006824	C-CER,CHIP;4700nF,10%,10V,X5R,TP,1608	4	SA	
...4	AD480	2203-006838	C-CER,CHIP;2200nF,10%,6.3V,X5R,TP,1005	12	SA	
...4	AD480	2203-006890	C-CER,CHIP;10000nF,20%,6.3V,X5R,TP,1608	32	SNA	
...4	AD480	2203-007176	C-CER,CHIP;10000nF,10%,16V,X5R,TP,2012,1	4	SNA	
...4	AD480	2203-007240	C-CER,CHIP;22000nF,20%,6.3V,X5R,TP,1810,	12	SA	
...4	AD480	2203-007269	C-CER,CHIP;22000nF,20%,10V,X5R,TP,2012(2	9	SA	
...4	AD480	2203-007270	C-CER,CHIP;10000nF,10%,10V,X5R,TP,1608,T	6	SNA	
...4	AD480	2203-007306	C-CER,CHIP;10000nF,10%,25V,X5R,TP,2012,1	7	SNA	
...4	AD480	2203-007486	C-CER,CHIP;1000nF,10%,50V,X5R,TP,1608	1	SA	
...4	AD480	2203-007544	C-CER,CHIP;100nF,10%,50V,X7R,TP,1005,T0.	9	SA	
...4	AD480	2203-008096	C-CER,CHIP;2200nF,10%,50V,X5R,TP,2012,1.	3	SA	
...4	AD480	2203-008315	C-CER,CHIP;22000nF,20%,25V,X5R,TP,2012,T	8	SA	
...4	AD480	2203-009740	C-CER,CHIP;1000nF,10%,100V,X7R,TP,3216,T	2	SA	
...4		2402-001268	C-AL,SMD;100uF,20%,25V,WT,TP,8x6.3mm	1	SA	
...4		2402-001276	C-AL,SMD;47uF,20%,35V,TP,6.6x6.6x5.8mm	1	SNA	
...4		2703-000213	INDUCTOR-SMD;470nH,10%,1.35Ohm,35mA,15,M	1	SA	
...4		2703-002269	INDUCTOR-SMD;56nH,5%,1005,T0.5,1.4Ohm,15	2	SA	
...4		2703-003149	INDUCTOR-SMD;2.2uH,20%,0.055Ohm,3000mA,W	1	SA	
...4		2703-003790	INDUCTOR-SMD;4.7uH,20%,8080,0.025Ohm,450	1	SA	

ANNEX. Exploded View & Part List

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
...4		2703-004575	INDUCTOR-SMD;22uH,20%,15x15mm,T8.5,0.058	1	SNA	
...4		2703-004724	INDUCTOR-SMD;8.2uH,20%,5050,T4,0.072Ohm,	4	SA	
...4		2703-005191	INDUCTOR-SMD;1.5uH,20%,6060,T4.5,0.020Ohm	1	SA	
...4		2703-005192	INDUCTOR-SMD;4.7uH,20%,6060,T4.5,0.036Oh	2	SA	
...4		2801-003773	CRYSTAL-SMD;12MHz,30ppm,28-AAN,20pF,50Oh	1	SA	
...4		3301-000314	BEAD-SMD;120ohm,1608,TP,120ohm/100MHz	1	SA	
...4		3301-001364	BEAD-SMD;1000ohm,1608,TP,1085ohm/108MHz,	6	SNA	
...4		3301-002039	BEAD-SMD;26ohm,1608,TP	25	SA	
...4		3301-002315	BEAD-SMD;120ohm,3216,TP	1	SNA	
...4		3404-001379	SWITCH-TACT;12V,50mA,250gf,4.9x4.9x1.5mm	1	SA	
...4		3601-001374	FUSE-SURFACE MOUNT;32V,5A,FAST-ACTING,PL	2	SA	
...4		3701-001967	CONNECTOR-HDMI;19P,A,FEMALE,AU,0.5mm,BLK	2	SA	
...4		3707-001106	CONNECTOR-OPTICAL;STRAIGHT,SPDIF,2.5PI	1	SA	
...4		3708-001150	CONNECTOR-FPC/FFC/PIC;30P,1mm,SMD-A,SN,Z	1	SA	
...4		3711-007776	CONNECTOR-HEADER;BOX,8P,1R,1.25mm,SMD-A,	1	SA	
...4	EH01	3711-008131	HEADER-BOARD TO CABLE;BOX,4P,1R,2.5mm,AN	1	SA	
...4	EH01	3711-008690	HEADER-BOARD TO CABLE;BOX,8P,1R,2.5mm,AN	1	SA	
...4		3711-009090	CONNECTOR-HEADER;BOX,16P,1R,1.25mm,SMD-A	1	SA	
...4		3722-003229	JACK-MODULAR;8P/8C,Y,STRAIGHT,NONE,AU,1P	1	SA	
...4		3722-003322	JACK-DC POWER;6P/4C,4PI,SN/PB,BLK,12.4x1	1	SA	
...4		3722-003457	JACK-USB;4P/1C,NI,BLK,ANGLE,A,2.0,13.1x1	1	SA	
...4	JACK PIN	3722-003546	JACK-PIN;5P,NI/SN,BLU/GRN/RED/WHT/YEL,SM	1	SA	
...4		6302-001376	GASKET-SMD;,,CONDUCTIVE FABRIC,T14,W43,L	6	SNA	
...4		BN40-00329A	TUNER-DTV AIR CABLE;DTVS20EH7A,VSF Half,	1	SA	
...4		BN41-02574B	PCB-MAIN;M5300,FR-4,4L,1.2T,1,Kant-S,144	1	SNA	
...4	CB07	BN61-13312B	BRACKET-SCREWLESS PCB;55KS8000,SK5,T0.3,	4	SNA	
...4		BN97-13036A	ASSY MICOM-MAIN;T-KTSAKUC-0609.10,1107-0	1	SNA	
...5		1107-002505	IC-EMMC;THGBMDG5D1LBAIT,4Gbyte,x1,WFBGA,	1	SA	
...4		BN97-13274A	ASSY MICOM-SUB;T-KTSINTV,UNV72,1107-0022	1	SNA	
...5		1107-002226	IC-NOR FLASH;W25Q40CLSSIP,4Mbit,SOIC,8P,	1	SA	
..3		BN97-13396A	ASSY DRM;KANT-S,ATSC,MAC,HDMP	1	SNA	
...4		BN46-00109H	KEY CODE-CERTIFICATION;MAC,TV/AV,General	1	SNA	
...4		BN46-00110P	KEY CODE-CERTIFICATION;MIRACAST(HDCP2.2)	1	SNA	
1		BN92-22112A	ASSY P/MATERIAL;UM4*	1	SNA	
0.2		6902-002669	BAG ROLL;HDPE/HDPE/PE FOAM,T0.015/T0.5,W	1	SNA	
0.2		6922-000013	BAND;PP,W18,L2300/L2900,TRP	0	SNA	
0.2		BN02-00319B	TAPE-SINGLE FACE;OPP,T0.05,W75,L800M,CLE	2	SNA	
0.2	AS080	BN63-07229A	SHEET-PET;rose70",PET,T0.052,W50,L100000	1	SNA	
0.2		BN69-16100A	CUSHION-SET;28M4500,EPS,16.7g/l,WHITE	1	SNA	
..3		BN81-01918A	A/S-RESIN;EPS,SG-302	0	SNA	
0.2		BN69-16936A	PALLET-WOODEN;28M4500,WOOD,1135,840,H120	1	SNA	
0.2		BN69-16940A	PAD;28M4500,CB,1125,830,YELLOW,CB-PE	1	SNA	
1		BN92-22113B	ASSY BOX;UM4*	1	SNA	
0.2		BN68-07893A	LABEL BOX;ALL,THERMAL PAPER,W60,L110,WHT	1	SNA	
0.2		BN69-16255A	BOX UNIT;28M4500,CB,SW4,A1,L810,W106,H44	1	SNA	
1		BN92-22114B	ASSY LABEL;UM4*	1	SNA	
0.2		BN02-00102B	TAPE-SINGLE FACE;FILAMENT,#8917,T0.15,W2	0	SNA	

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
0.2		BN68-06708G	LABEL-RATING;Monitor,WW,PP,T0.161,W93,L7	1	SNA	
0.2		BN68-08562F	LEAFLET-QUICK SETUP GUIDE;UA32M5500AWXXY	1	SNA	
0.2		BN68-08653D	LABEL-ENERGY;UN28M4500AFXZA,UNITED STATE	1	SNA	
1	ACCE1	BN92-22115A	ASSY ACCESSORY;UM4*	1	SNA	
0.2	P001A	BN44-00835A	DC VSS(A);A4819_FDY,19V,2.53A,100-240V,5	1	SA	
0.2		BN96-43806C	ASSY ACCESSORY-MANUAL CABLE;UM4* 28,ZA	1	SNA	
..3	T0268	3903-001117	CBF-POWER CORD;DT,USA,Angle,2P(C7),125V,	1	SA	
..3		4301-000121	BATTERY-MN;1.5V,R03,10.5x44.5m,7.5g,AAA	2	SNA	
..3		6902-001962	BAG PE;LDPE,BIOBASED,T0.05,W400,L300,TRP	1	SNA	
..3		BN59-01267A	REMOCON-TV;2017 TV,Samsung,45KEY,3V,Nort	1	SA	
..3	HOLDER-WIRE	BN61-08370A	HOLDER-STAND CABLE;26UE4000,PA66	1	SA	
...4		0103-010724	RESIN PA;A3K Q790,Uncolored	3	SNA	
..3	T0527	BN68-00513A	LABEL-E PASS;ALL MODEL,WW,YUPO,W50,L15,W	1	SNA	
..3		BN68-04972E	LEAFLET-REGULATORY GUIDE;ALL,SAMSUNG,WW	1	SNA	
..3		BN68-08058H	MANUAL USERS;UA43M5500AWXXY,AUSTRALIA,AU	1	SNA	
1		BN95-04265A	PRODUCT LCD-BOE;CY-HM028AGEV1V/H,M4500,B	1	SA	
0.2		BN68-05722A	LABEL-E PASS;POLYPROPYLENE,NON-COATING	1	SNA	
0.2		BN74-00053G	TAPE-SINGLE FACE;PAPER,3M2307,T0.14,W10,	0	SNA	
0.2		BN90-09149A	ASSY BACK LIGHT UNIT;28M41,45	1	SNA	
..3		BN61-15275A	OPTICAL SHEET-REFLECTOR;17Y_M4500_28INCH	1	SNA	
..3		BN61-15276A	OPTICAL SHEET-COMPLEX;17Y_M4500_28INCH_C	1	SNA	
..3		BN61-15277A	DIFFUSER PLATE;17Y_M4500_28INCH_D.P,PS,T	1	SNA	
..3		BN96-44316A	ASSY CHASSIS REAR P;28M4500,EGI-SECC,WIT	1	SNA	
...4		BN96-43700A	ASSY CHASSIS REAR P;28M4500,EGI-SECC,0.4	1	SNA	
....5		BN02-00040C	TAPE DOUBLE FACE;F-LED ALL,PET,T0.3,W3,L	2	SNA	
....5		BN02-00324A	TAPE-SINGLE FACE;PAPER,Y14 4K,T0.085,W25	0	SNA	
....5		BN02-00352B	TAPE-SINGLE FACE;PET,T0.05,W30,L50M,BLAC	0	SNA	
....5		BN02-00352D	TAPE-SINGLE FACE;PET,T0.05,W20,L50M,BLAC	0	SNA	
....5		BN61-09552B	HOLDER-SOURCE PCB;SHARP HD R,ABS,BLACK,H	1	SNA	
....6		0103-004609	RESIN ABS;HF-0680U,K21294,BK0007,HB,High	2	SNA	
....5		BN61-09564A	HOLDER-SOURCE PCB;Y13 SF-LED(HD) 32",ABS	1	SNA	
....5		BN61-15021A	FRAME-MIDDLE TOP LEFT;28M4500,TPV,HB	1	SNA	
....5		BN61-15022A	FRAME-MIDDLE TOP RIGHT;28M4500,TPV,HB	1	SNA	
....5		BN61-15023A	FRAME-MIDDLE BOTTOM LEFT;28M4500,TPV,HB	1	SNA	
....5		BN61-15077A	FRAME-MIDDLE BOTTOM RIGHT;28M4500,TPV,HB	1	SNA	
....6		0103-010669	RESIN TPE;5402PV/5409H,Black,BK0008,HB,T	9	SNA	
....5	M0131	BN63-00135A	GASKET-EMI;MCX15A,Conductive Fabric,T2.5	2	SNA	
....5	CC04	BN64-03882A	CHASSIS-REAR;28M4500,EGI-SECC,PRESS,T0.4	1	SNA	
....6		BN01-00513A	STEEL;28M4500,EGI-SECC,L695,0.45mm,435mm	1	SNA	
....6	CB20	BN61-13521A	BRACKET-WALL;32K5500,CR-SPCC,T1.2,NATURA	4	SNA	
....6		BN61-13631A	STUD-PEM;55KS8000,SUM24L,T0.85,W4,L11.2,	2	SNA	
....5		BN96-30413A	ASSY PCB P-LED INTERFACE BOARD;28H4000,-	1	SNA	
....5		BN96-30440A	ASSY LED BAR P;28inch HD Direc,CEM3,FCPk	3	SNA	
0.2		BN96-30087A	ASSY OPEN CELL;BOE,28Inch,Y2014 New Mode	1	SA	
..3		BN81-07791A	A/S-ADHESIVE ACF;HV320WX2-261,42-706007,	1	SNA	
..3	M0014	BN81-07792A	A/S-PCB;PCB-ACF,42-706012,ACF(AC90033R45	1	SNA	

ANNEX. Exploded View & Part List

Level	Location No.	Code No.	Description & Specification	Q'ty	SA/SNA	Remark
..3		BN81-09545A	A/S-POLARIZER TFT;HV280WX2-260,38-100116	1	SNA	
..3		BN81-09546A	A/S-POLARIZER CF;HV280WX2-260,38-1001164	1	SNA	
..3		BN81-09547A	A/S-ASSY PCB SOURCE;HV280WX2-260,44-9711	1	SNA	
..3		BN81-09548A	A/S-IC DRIVER GATE;HV280WX2-260,47-70210	1	SNA	
..3		BN81-09549A	A/S-IC DRIVER SOURCE;HV280WX2-260,47-701	1	SNA	
0.2		BN96-43701A	ASSY CHASSIS FRONT P;28M4500,PC+ABS+ED18	1	SA	
..3		BN60-00667A	SPACER-CONDUCTIVE;Y13 SF-LED,CONDUCTIVE	4	SNA	
..3		BN60-00715A	SPACER-CONDUCTIVE;Y13 Slim F-LED,CONDUCT	4	SNA	
..3		BN60-01366L	SPACER-CONDUCTIVE;CONDUCTIVE FABRIC/PU F	2	SNA	
..3		BN60-01580A	SPACER-CONDUCTIVE;CONDUCTIVE FABRIC/PU F	2	SNA	
..3	AS080	BN63-07556N	SHEET-PROTECTION COVER;TOC,PO,W40,L10000	1	SNA	
..3	AS080	BN63-07556Y	SHEET-PROTECTION COVER;Y13 SF-LED,PO,T0.	2	SNA	
..3	AC155	BN64-03881A	CHASSIS-FRONT;28M4500,PC+ABS+ED18%,V-1,-	1	SNA	
...4		0103-010808	RESIN PC ABS;NH-1210/K21294,BLACK,BK0007	170	SNA	